

Building and evaluation of a PBPK model for Rosuvastatin in healthy adults

Version	master-OSP12.2
based on <i>Model Snapshot and Evaluation Plan</i>	https://github.com/Open-Systems-Pharmacology/Rosuvastatin-Model/releases/tag/vmaster
OSP Version	12.2
Qualification Framework Version	3.5

This evaluation report and the corresponding PK-Sim project file are filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/>

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1 Introduction

Rosuvastatin is an inhibitor of the liver 3-hydroxy-3-methyl-glutaryl-coenzyme A reductase (HMG-CoA reductase) and is prescribed to treat hyperlipidemia. Rosuvastatin is metabolized by CYP2C9 to a small extent, transported by OATP1B1 and OATP1B3 into the liver, and subsequently excreted into the bile by BCRP. The renal excretion of rosuvastatin is high, suggesting active tubular secretion (by the transporters OATP2B1, OAT3, and Pgp). The absolute bioavailability of rosuvastatin is about 20%, mainly due to poor absorption. The fraction excreted in feces and urine after an oral dose is about 90% and 10%, respectively, both mainly as parent compound. Rosuvastatin is a substrate of OATP1B1, OATP1B3, and BCRP and can, therefore, serve as a victim for these transporters in DDI studies.

The herein presented PBPK model of rosuvastatin has been developed using published pharmacokinetic clinical data by Billington et al. 2019 ([Billington 2019](#)), Martin et al. 2003 ([Martin 2003c](#)), Stopfer et al. 2016 ([Stopfer 2016](#)), Stopfer et al. 2018 ([Stopfer 2018b](#)), Wu et al. 2017 ([Wu 2017](#)), Gidal et al. 2017 ([[Gidal 2017](#)])(#5-references)), Cooper et al. 2003 ([Cooper 2003a](#)), Lee et al. 2018 ([Lee 2018](#)) and Gosai et al. 2008 ([Gosai 2008](#)). The model has then been evaluated by comparing observed data to simulations of both intravenously and orally administered rosuvastatin covering a dose range of 0.002 mg to 80.0 mg. Both single dose and multiple dose administration are represented in the development and evaluation data sets.

The presented model includes the following features:

- metabolism by CYP2C9,
- transport by OATP1B1 and OATP1B3 (lumped together for identifiability),
- transport by BCRP,
- transport by OATP2B1,
- transport by Pgp,
- transport by OAT3,
- renal clearance by glomerular filtration,
- oral absorption with dissolution rate described by a Weibull function.

2 Methods

2.1 Modeling strategy

The general concept of building a PBPK model has previously been described by Kuepfer et al. ([Kuepfer 2016](#)). Relevant information on anthropometric (height, weight) and physiological parameters (e.g. blood flows, organ volumes, binding protein concentrations, hematocrit, cardiac output) in adults was gathered from the literature and has been previously published ([Willmann 2007](#)). The information was incorporated into PK-Sim® and was used as default values for the simulations in adults.

The applied activity and variability of plasma proteins and active processes that are integrated into PK-Sim® are described in the publicly available PK-Sim® Ontogeny Database Version 7.3 ([PK-Sim Ontogeny Database Version 7.3](#)) or otherwise referenced for the specific process.

A mean model was built based on clinical data from studies with intravenous ([Billington et al. 2019](#) ([Billington 2019](#)), [Martin et al. 2003](#) ([Martin 2003c](#))) and oral administration ([Martin et al. 2003](#) ([Martin 2003c](#)), [Stopfer et al. 2016](#) ([Stopfer 2016](#)), [Stopfer et al. 2018](#) ([Stopfer 2018b](#)), [Wu et al. 2017](#) ([Wu 2017](#)), [Gidal et al. 2017](#) ([Gidal 2017](#)), [Cooper et al. 2003](#) ([Cooper 2003a](#)), [Lee et al. 2018](#) ([Lee 2018](#)) and [Gosai et al. 2008](#) ([Gosai 2008](#))) of rosuvastatin. The studies reported individual ([Billington 2019](#)) or mean plasma concentrations of rosuvastatin. The study by [Billington et al. 2019](#) also reported individual (n=1) bile fraction of unchanged rosuvastatin as well as tissue concentration in liver and kidney, and the studies by [Martin et al. 2003](#), [Stopfer et al. 2016](#) and [Stopfer et al. 2018](#) reported fraction excreted in urine of unchanged rosuvastatin in addition to plasma concentrations. The mean PBPK model was developed using a mean individual based on the demographic data for each study. If no demographic data were provided, the following values were used: male, European, 30 years of age, 73 kg body weight, and 176 cm body height. The relative tissue-specific expressions of the enzyme and transporter predominantly being involved in the metabolism/transport of rosuvastatin (CYP2C9, OATP1B1/, BCRP, Pgp and OAT3) were considered ([Meyer 2012](#)). A Weibull function was used to describe the oral dissolution of rosuvastatin and the dose was applied as a split dose approach with lag time for the second dose to capture the late T_{max}.

A specific set of parameters (see below) was optimized to describe the disposition of rosuvastatin using the Parameter Identification module provided in PK-Sim®. Structural model selection was mainly guided by visual inspection, mean relative deviation and geometric mean fold error of all predicted AUC_{last} and C_{max} values.

The model was then verified by simulating further clinical studies reporting pharmacokinetic concentration-time profiles after intravenous and oral administration of rosuvastatin.

Details about input data (physicochemical, *in vitro* and clinical) can be found in [Section 2.2](#).

Details about the structural model and its parameters can be found in [Section 2.3](#).

2.2 Data used

2.2.1 In vitro and physicochemical data

A literature search was performed to collect available information on physicochemical properties of rosuvastatin. The obtained information from literature is summarized in the table below, and is used for model building. Final model parameters are stated in [Section 3.1](#).

Parameter	Unit	Value	Source	Description
MW	g/mol	481.54	Wishart 2006	Molecular weight
pK _a (acid)		4.3	Riccardi 2019	acid dissociation constant of conjugate acid
Solubility (water)	g/L	7.8	TGA 2011	Aqueous solubility
logP		-0.33	McTaggart 2001 and Jones 2012	Partition coefficient between octanol and water
fu	%	11.5	FDA 2003	Fraction unbound in plasma
K _m BCRP	μmol/L	2.02	Kitamura 2008	BCRP Michaelis-Menten constant
K _m OAT3	μmol/L	7.40	Windass 2007	OAT3 Michaelis-Menten constant
K _m OATP2B1	μmol/L	6.42	Kitamura 2008	OATP2B1 Michaelis-Menten constant
K _m OATP1B1/3	μmol/L	0.80	Kitamura 2008	OATP1B1/3 Michaelis-Menten constant
K _m Pgp	μmol/L	203	Goard 2010	Pgp Michaelis-Menten constant
Weibull shape	-	1.77	Seo 2019	Dissolution profile shape
Weibull time	min	6.25	Seo 2019	Dissolution time (50% dissolved)

2.2.2 Clinical data

A literature search was performed to collect available clinical data on rosuvastatin in adults.

The following publications were found in adults for model building:

Publication	Arm / Treatment / Information used for model building
Billington 2019	Individual arterial plasma and whole blood, and tissue liver and kidney PK profiles and fraction excreted to bile in healthy subjects after single intravenous administration of 0.00174 mg rosuvastatin 160 minutes after an oral dose of 5 mg rosuvastatin
Martin 2003c	Plasma PK profiles and urine data in healthy subjects after single intravenous infusion (4 h) of 8 mg rosuvastatin and single oral administration of 40 mg rosuvastatin tablet
Stopfer 2016	Plasma PK profiles and urine data in healthy subjects after single oral administration of 10 mg rosuvastatin tablet
Stopfer 2018b	Plasma PK profiles and urine data in healthy subjects after single oral administration of 10 mg rosuvastatin tablet
Wu 2017	Plasma PK profiles in healthy subjects after single oral administration of 20 mg rosuvastatin tablet
Gidal 2017	Plasma PK profiles in healthy subjects after single oral administration of 40 mg rosuvastatin tablet
Cooper 2003a	Plasma PK profiles in healthy subjects after single oral administration of 80 mg rosuvastatin tablet
Lee 2018	Plasma PK profiles in healthy subjects after multiple oral administration of 20 mg rosuvastatin QD
Gosai 2008	Plasma PK profiles in healthy subjects after multiple oral administration of 40 mg rosuvastatin QD

The following table shows the data from the excretion studies used for model building:

Observer	Value
Fraction excreted to bile of unchanged rosuvastatin at 0.46 h after iv administration of 0.00174 mg	4.76%
Fraction excreted to urine of unchanged rosuvastatin after iv infusion of 8 mg	29.5%
Fraction excreted to urine of unchanged rosuvastatin after oral administration 10 mg	6.27% and 5.1%
Fraction excreted to urine of unchanged rosuvastatin after oral administration 40 mg	5.09%

The following dosing scenarios were simulated and compared to respective data for model verification:

Scenario	Data reference
po 0.05	Prueksaritanont 2017
po 5	Prueksaritanont 2014
po 10 mg	Cooper 2003b , Csonka 2019 , Huguet 2016 , Martin 2003b , Stopfer 2018a , Wiebe 2020 and Coss 2016
po 20 mg	Martin 2003d , FDA 2003 , Edwards 2017 , Martin 2003b , Birmingham 2015 , Jones 2020 , Lee 2018 , Martin 2016 and Willis 2020
po 40 mg	Martin 2002a , Martin 2003b and Lee 2005
po 80 mg	Cooper 2002 , Cooper 2003b , Cooper 2003c , Martin 2003b , Schneck 2004
po 10 mg QD for 14 days	Kosoglou 2004 and Martin 2002b
po 20 mg QD for 7 days	FDA 2003 and FDA 2018

2.3 Model parameters and assumptions

2.3.1 Absorption

The parameter value for `Specific intestinal permeability` was optimized based on clinical oral data, see results of optimization in [Section 2.3.4](#). The measured solubility in water was used in the model (see [Section 2.2.1](#))

The dissolution of tablets was implemented via empirical Weibull dissolution extracted from literature. The dose was applied as a split dose approach with lag time for the second dose to capture the late Tmax as informed by popPK modeling.

2.3.2 Distribution

Rosuvastatin is highly bound to plasma proteins (approx. 88.5 %) (see [Section 2.2.1](#)). A value of 11.5% was used in this PBPK model for `Fraction unbound (plasma, reference value)`. The major binding partner was set to albumin (see [Section 2.2.1](#)).

An important parameter influencing the resulting volume of distribution is lipophilicity. The reported experimental logP value of -0.33 was used in this model (see [Section 2.2.1](#)).

After testing the available organ-plasma partition coefficient and cell permeability calculation methods built in PK-Sim, observed clinical data was best described by choosing the partition coefficient calculation by `PK-Sim Standard` and cellular permeability calculation by `PK-Sim Standard`.

2.3.3 Metabolism and Elimination

One metabolic pathway was implement into the model via Michaelis-Menten kinetics

- CYP2C9

The CYP2C9 expression profiles is based on high-sensitive real-time RT-PCR ([Nishimura 2003](#)). Metabolic enzyme activity was described as saturable process following Michaelis-Menten kinetics, where the `Km` was taken from literature and the `kcat` was optimized based on clinical data (see [Section 2.3.4](#)).

The following transport proteins were implemented into the model with the Michaelis-Menten kinetics

- OAT3

The OAT3 expression profiles is based on high-sensitive real-time RT-PCR ([Nishimura 2003](#)). Transporter activity was described as saturable process following Michaelis-Menten kinetics, where the `Km` was set to 1 $\mu\text{mol/l}$ and `kcat` was optimized based on clinical data (see [Section 2.3.4](#)).

- BCRP

The BCRP expression profiles is based on whole genome expression arrays from ArrayExpress ([Kolesnikov 2015](#)). Transporter activity was described as saturable process following Michaelis-Menten kinetics, where the `Km` was taken from literature and `kcat` was optimized based on clinical data (see [Section 2.3.4](#)).

- Pgp

The Pgp expression profiles is based on high-sensitive real-time RT-PCR ([Nishimura 2003](#)) with an intestinal mucosa of factor 3.57 ([Hanke 2018](#)). Transporter activity was described as saturable process following Michaelis-Menten kinetics, where the `Km` was taken from literature and `kcat` was optimized based on clinical data (see [Section 2.3.4](#)).

- OATP1B1/3

For OATP1B1/3 the expression profiles was considered only for OATP1B1 and is based on high-sensitive real-time RT-PCR ([Nishimura 2003](#)). The reference concentration for OATP1B1 was measured by liquid chromatography tandem mass spectroscopy ([Prasad 2014](#)). Transporter activity was described as saturable process following Michaelis-Menten kinetics, where the `Km` was taken from literature and `kcat` was optimized based on clinical data (see [Section 2.3.4](#)).

- OATP2B1

The OATP2B1 expression profiles is based on whole genome expression arrays from ArrayExpress ([Kolesnikov 2015](#)). Transporter activity was described as saturable process following Michaelis-Menten kinetics, where the `Km` was taken from literature and `kcat` was optimized based on clinical data (see [Section 2.3.4](#)).

Additionally, passive renal clearance by glomerular filtration was implemented and the `GFR fraction` was set to 1. In addition, fraction of bile that was continuously released was set to 1 (`EHC continuous fraction`).

2.3.4 Automated Parameter Identification

The following parameters were optimized by fitting the model to the data:

Model Parameter
<code>kcat</code> (BCRP)
<code>kcat</code> (CYP2C9)
<code>kcat</code> (OATP1B1/3)
<code>kcat</code> (Pgp)
<code>kcat</code> (OAT3)
<code>kcat</code> (OATP2B1)
Specific intestinal permeability
<code>P(intracell->interstitial)</code> small intestine
<code>P(intracell->interstitial)</code> small intestine

3 Results and Discussion

The PBPK model for rosuvastatin was developed and verified with clinical pharmacokinetic data.

The model was built and evaluated covering data from studies including in particular

- single intravenous administration and both single and multiple oral administrations (tablets, capsules and solution).
- a dose range of 0.002 to 80.0 mg

The model quantifies excretion via urine and bile (by glomerular filtration and active transport) and metabolism via CYP2C9. The active transport is attributed to Pgp, BCRP, OATP1B1/3, OAT3 and OATP2B1.

The next sections show:

1. the final model input parameters for the building blocks: [Section 3.1](#).
2. the overall goodness of fit: [Section 3.2](#).
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: [Section 3.3](#).

3.1 Rosuvastatin final input parameters

The compound parameter values of the final PBPK model are illustrated below.

Compound: Rosuvastatin

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	7800 mg/l	Publication-In Vitro-TGA 2011	Measurement	True
Reference pH	7		Measurement	True
Lipophilicity	-0.33 Log Units	Publication-In Vitro-McTaggart 2001 and Jones 2012	Measurement	True
Fraction unbound (plasma, reference value)	0.115	Publication-In Vitro-FDA 2003	Measurement	True
Specific intestinal permeability (transcellular)	1.8086332716E-06 cm/min	Parameter Identification-Parameter Identification-Value updated from '29 - P large int, CYP2C9 - CoV - 5' on 2020-10-30 16:15	Fit	True
F	1			
Is small molecule	Yes			
Molecular weight	481.538 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	PK-Sim Standard
Cellular permeabilities	PK-Sim Standard

Processes

Transport Protein: BCRP-Paper

Molecule: BCRP

Parameters

Name	Value	Value Origin
In vitro Vmax/transporter	0 nmol/min/pmol transporter	
Km	2.02 µmol/l	Publication-In Vitro-Kitamura 2008
kcat	1.0072328564 1/min	Parameter Identification-Parameter Identification-Value updated from '29 - P large int, CYP2C9 - CoV - 5' on 2020-10-30 16:14

Systemic Process: Glomerular Filtration-GFR

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Parameter Identification-Parameter Identification-Value updated from '29 - P large int, CYP2C9 - CoV' on 2020-10-27 16:22

Transport Protein: OAT3-Paper

Molecule: OAT3

Parameters

Name	Value	Value Origin
In vitro Vmax/transporter	0 nmol/min/pmol transporter	
Km	7.4 µmol/l	Publication-In Vitro-Windass 2007
kcat	30999.6809549078 1/min	Parameter Identification-Parameter Identification-Value updated from '29 - P large int, CYP2C9 - CoV - 5' on 2020-10-30 16:15

Transport Protein: OATP1B1-Paper

Molecule: OATP1B1

Parameters

Name	Value	Value Origin
In vitro Vmax/transporter	0 nmol/min/pmol transporter	
Km	0.802 $\mu\text{mol/l}$	Publication-In Vitro-Kitamura 2008
kcat	805.2413910931 1/min	Parameter Identification-Parameter Identification-Value updated from '29 - P large int, CYP2C9 - CoV - 5' on 2020-10-30 16:15

Transport Protein: OATP2B1-Paper

Molecule: OATP2B1

Parameters

Name	Value	Value Origin
In vitro Vmax/transporter	0 nmol/min/pmol transporter	
Km	6.42 $\mu\text{mol/l}$	Publication-In Vitro-Kitamura 2008
kcat	4.3108123152 1/min	Parameter Identification-Parameter Identification-Value updated from '29 - P large int, CYP2C9 - CoV - 5' on 2020-10-30 16:15

Transport Protein: P-gp-Paper

Molecule: P-gp

Parameters

Name	Value	Value Origin
In vitro Vmax/transporter	0 nmol/min/pmol transporter	
Km	203 $\mu\text{mol/l}$	Publication-In Vitro-Goard 2010
kcat	46.0938220893 1/min	Parameter Identification-Parameter Identification-Value updated from '29 - P large int, CYP2C9 - CoV - 5' on 2020-10-30 16:14

Metabolizing Enzyme: CYP2C9-Hypothetical Km

Molecule: CYP2C9

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	0 nmol/min/pmol rec. enzyme	
Km	1 µmol/l	Other-Assumption-Assumed
kcat	0.00097440861845 1/min	Parameter Identification-Parameter Identification-Value updated from '29 - P large int, CYP2C9 - CoV - 5' on 2020-10-30 16:15

3.2 Rosuvastatin Diagnostics Plots

Below you find the goodness-of-fit visual diagnostic plots for the PBPK model performance of all data used presented in [Section 2.2.2](#).

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

Table 3-1: GMFE for Goodness of fit plot for concentration in plasma.

Group	GMFE
Rosuvastatin iv (model building)	1.49
Rosuvastatin oral administration (model building)	1.26
Rosuvastatin oral administration (model verification)	1.34
All	1.33

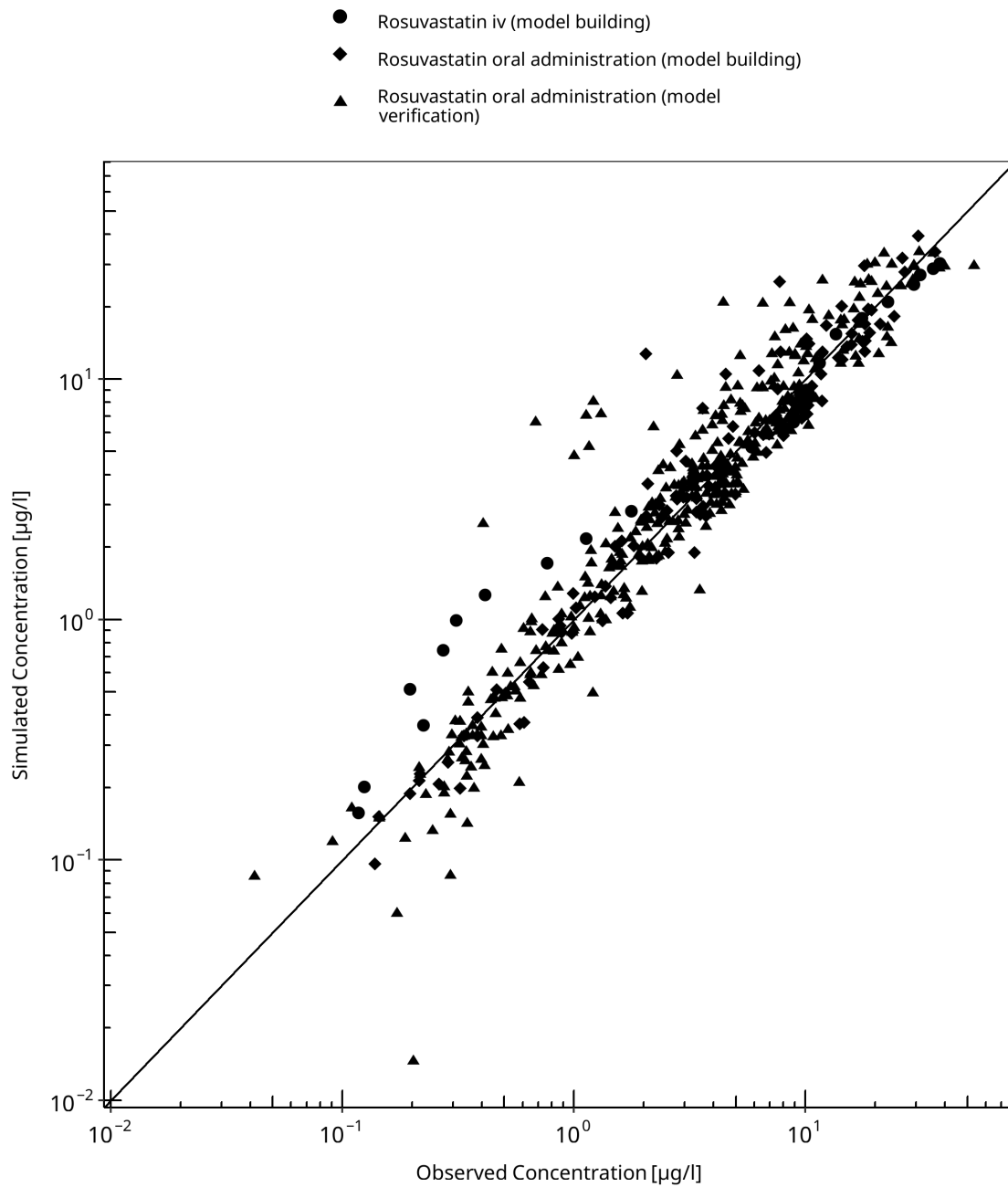


Figure 3-1: Goodness of fit plot for concentration in plasma.

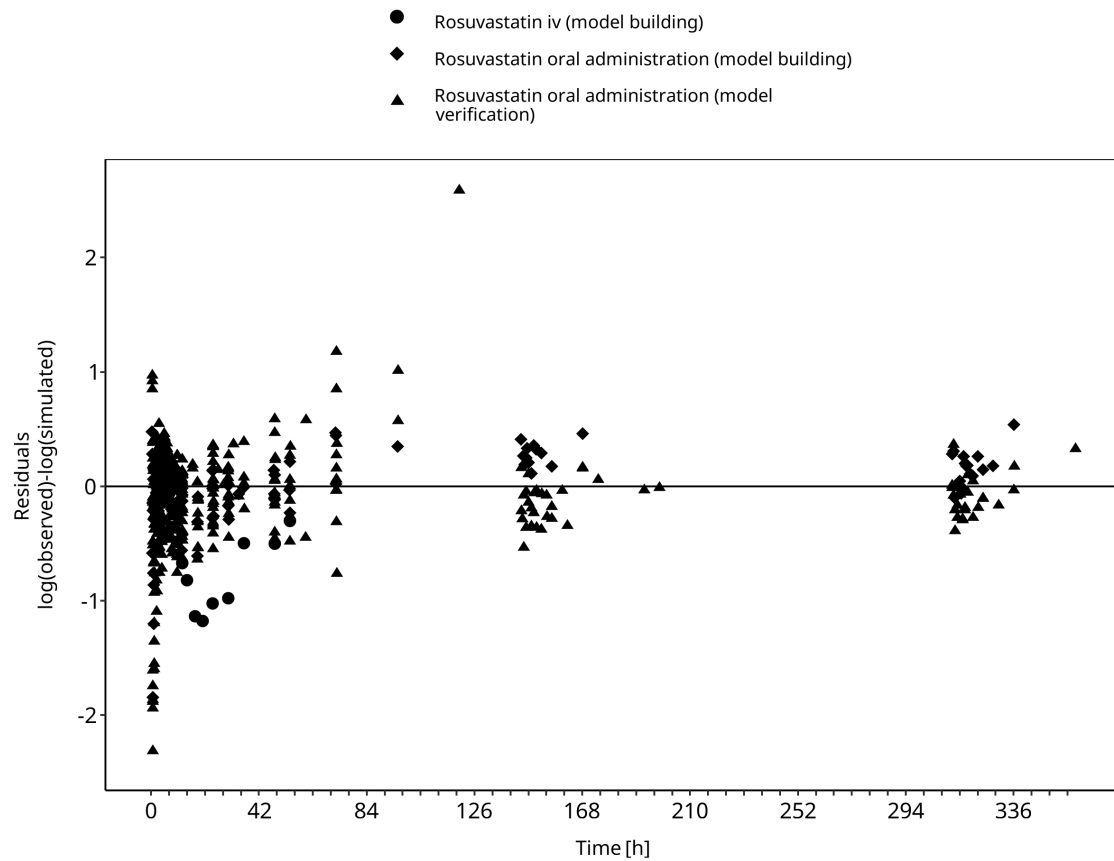


Figure 3-2: Goodness of fit plot for concentration in plasma.

3.3 Concentration-Time Profiles

Simulated versus observed concentration-time profiles of all data listed in [Section 2.2.2](#) are presented below.

3.3.1 Model Building

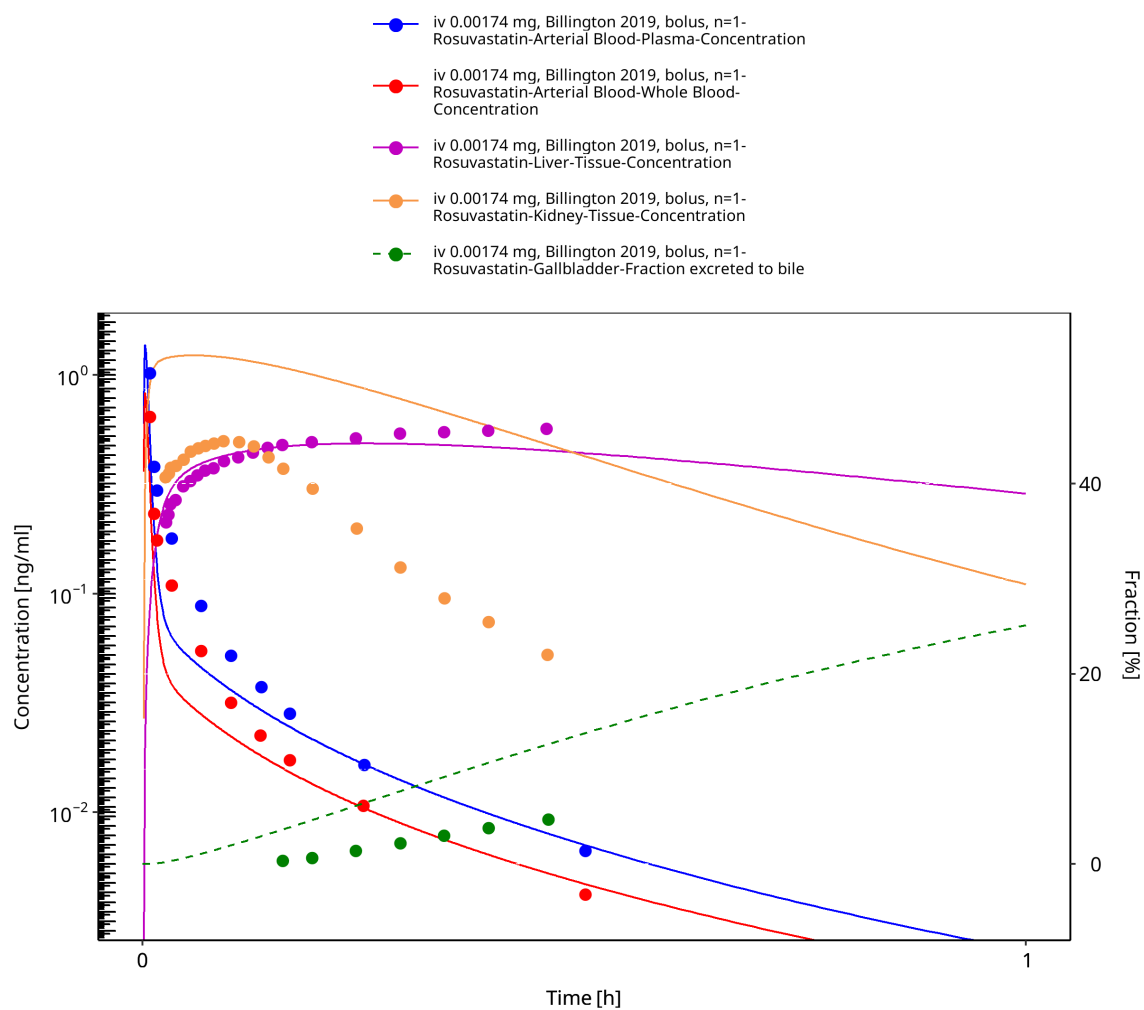


Figure 3-3: Billington 2019 - IV bolus 0.00174 mg control

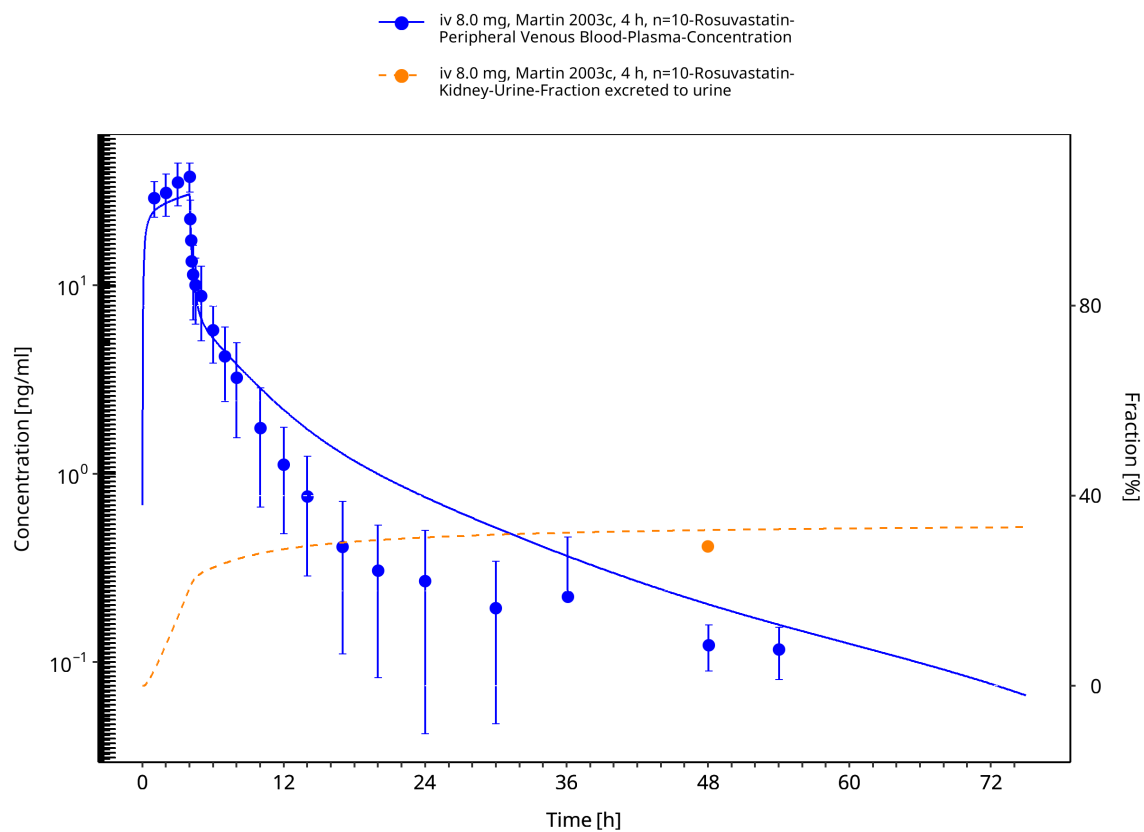


Figure 3-4: Martin 2003c - IV 8 mg control

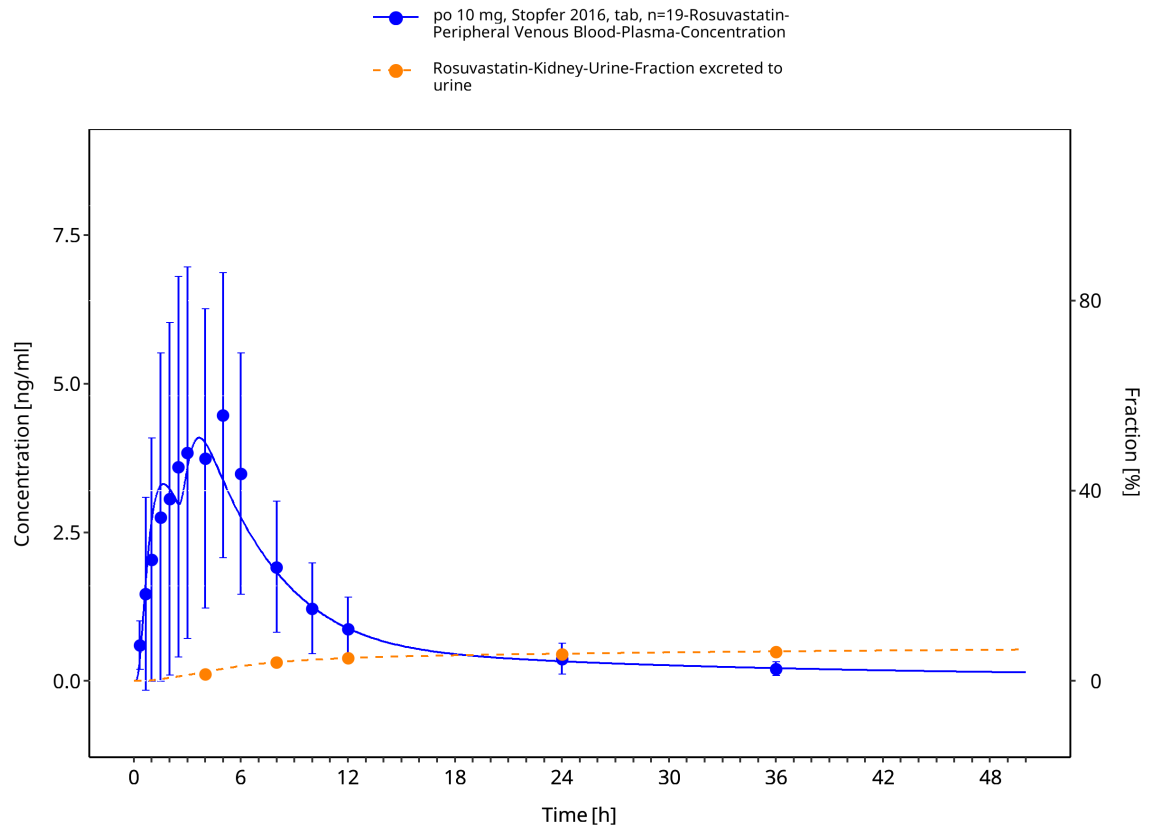


Figure 3-5: Stopfer 2016 - PO Rosuvastatin 10 mg control

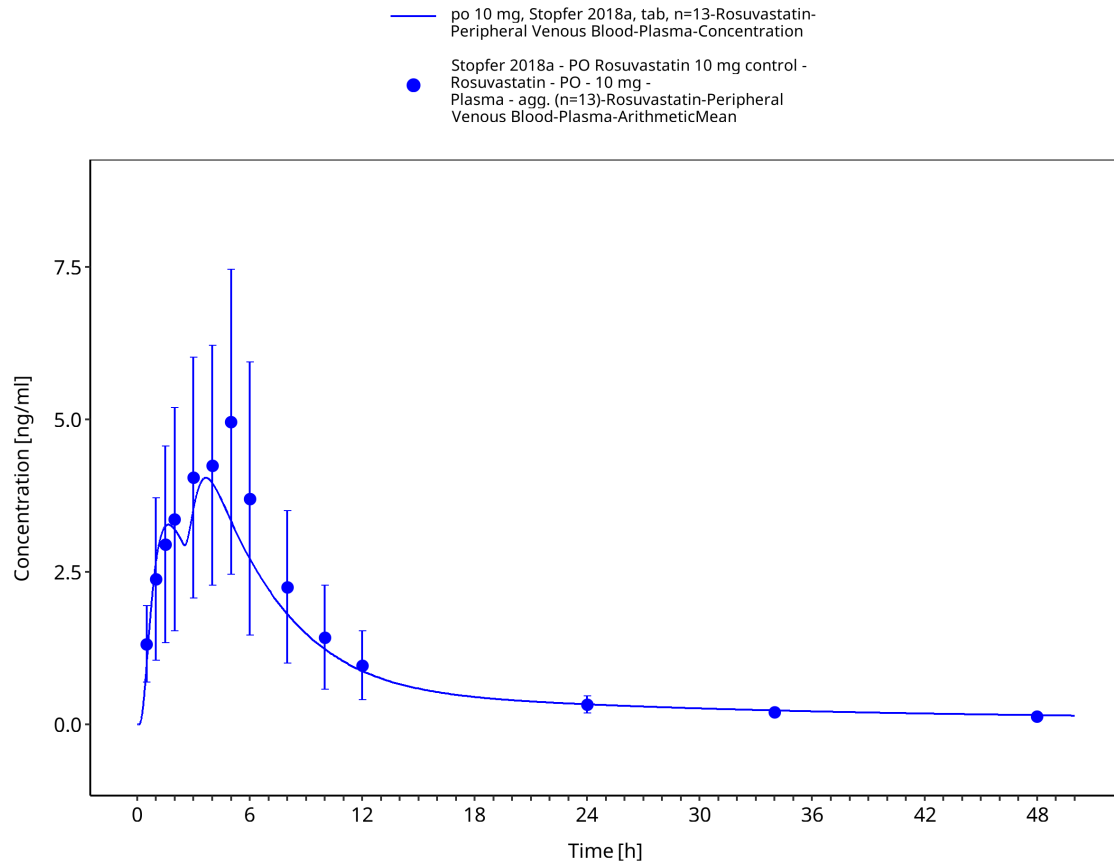


Figure 3-6: po 10 mg, Stopfer 2018a, tab

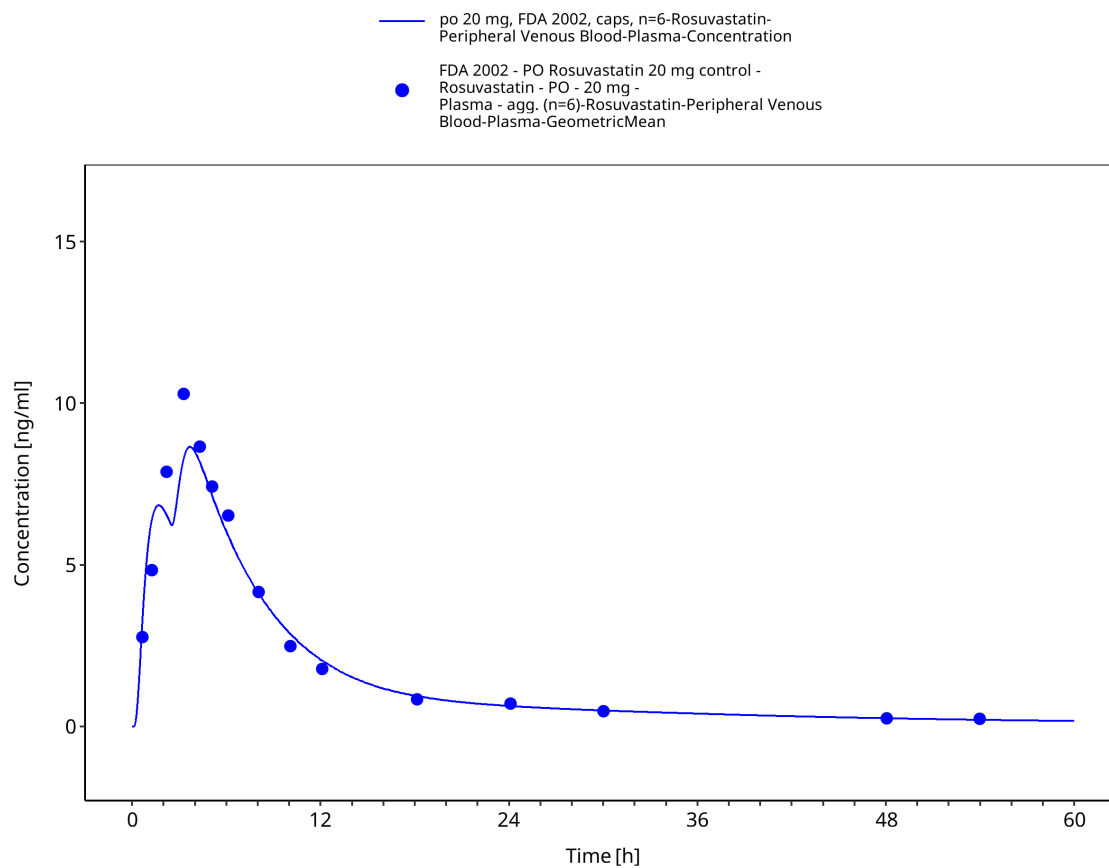


Figure 3-7: FDA 2002 - PO Rosuvastatin 20 mg control

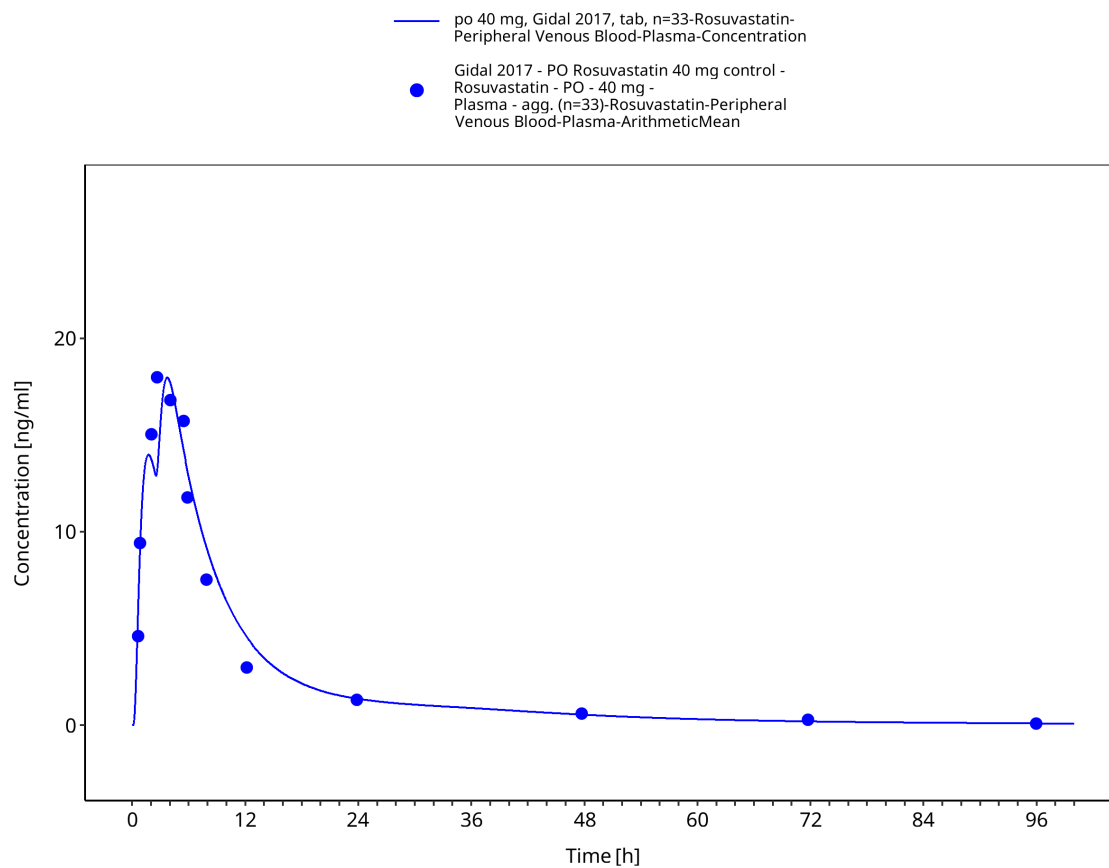


Figure 3-8: Gidal 2017 - PO Rosuvastatin 40 mg control

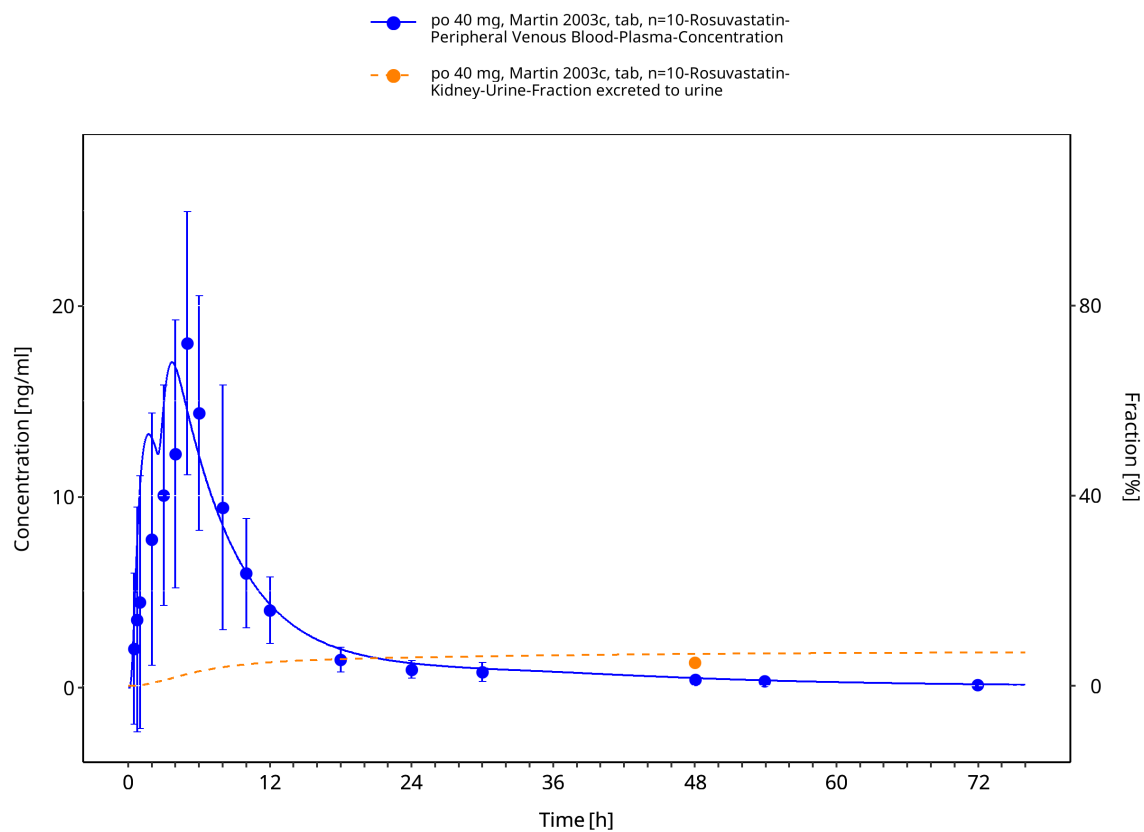


Figure 3-9: Martin 2003c - PO Rosuvastatin 40 mg control

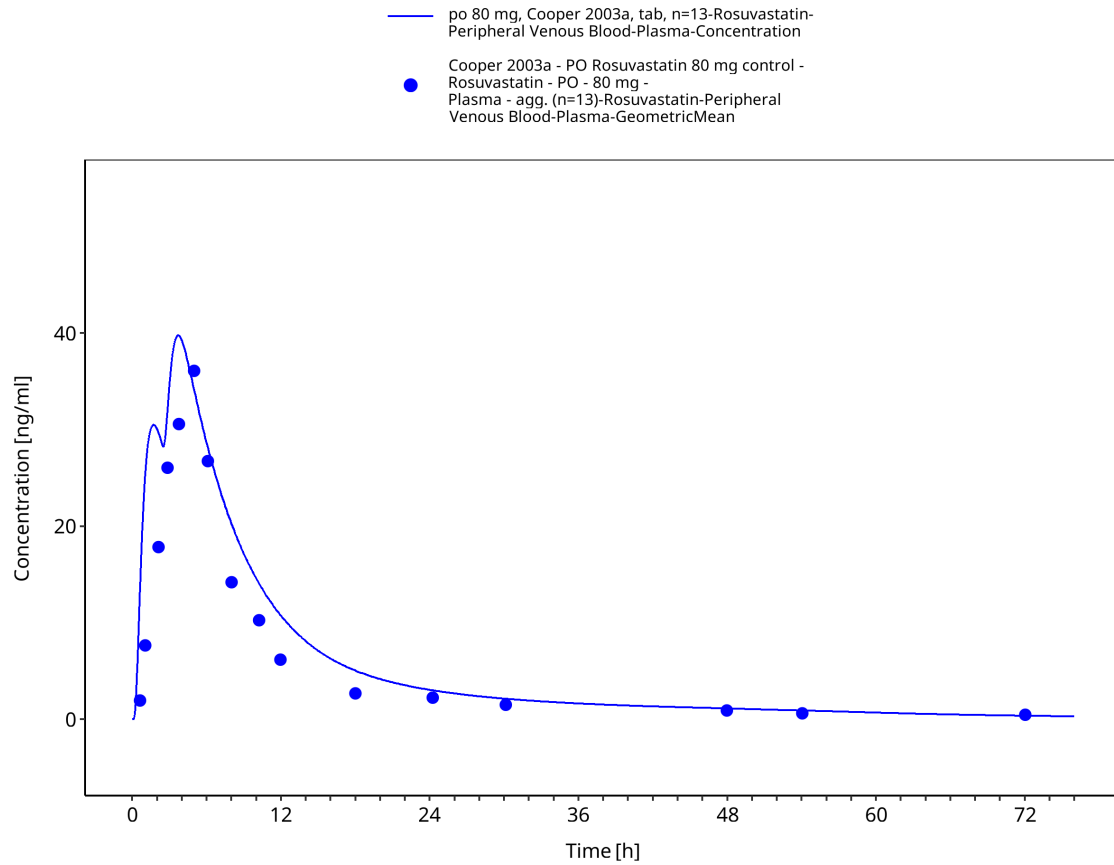


Figure 3-10: Cooper 2003a - PO Rosuvastatin 80 mg control

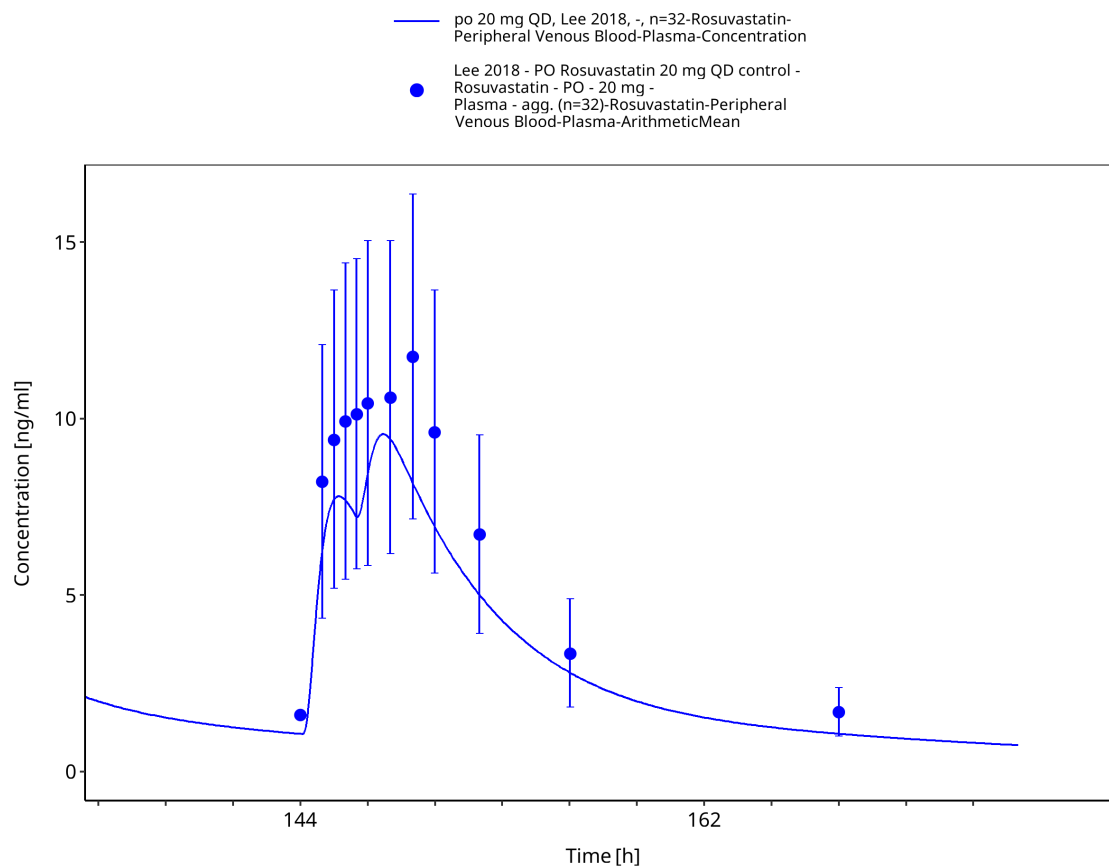


Figure 3-11: Lee 2018 - PO Rosuvastatin 20 mg QD control

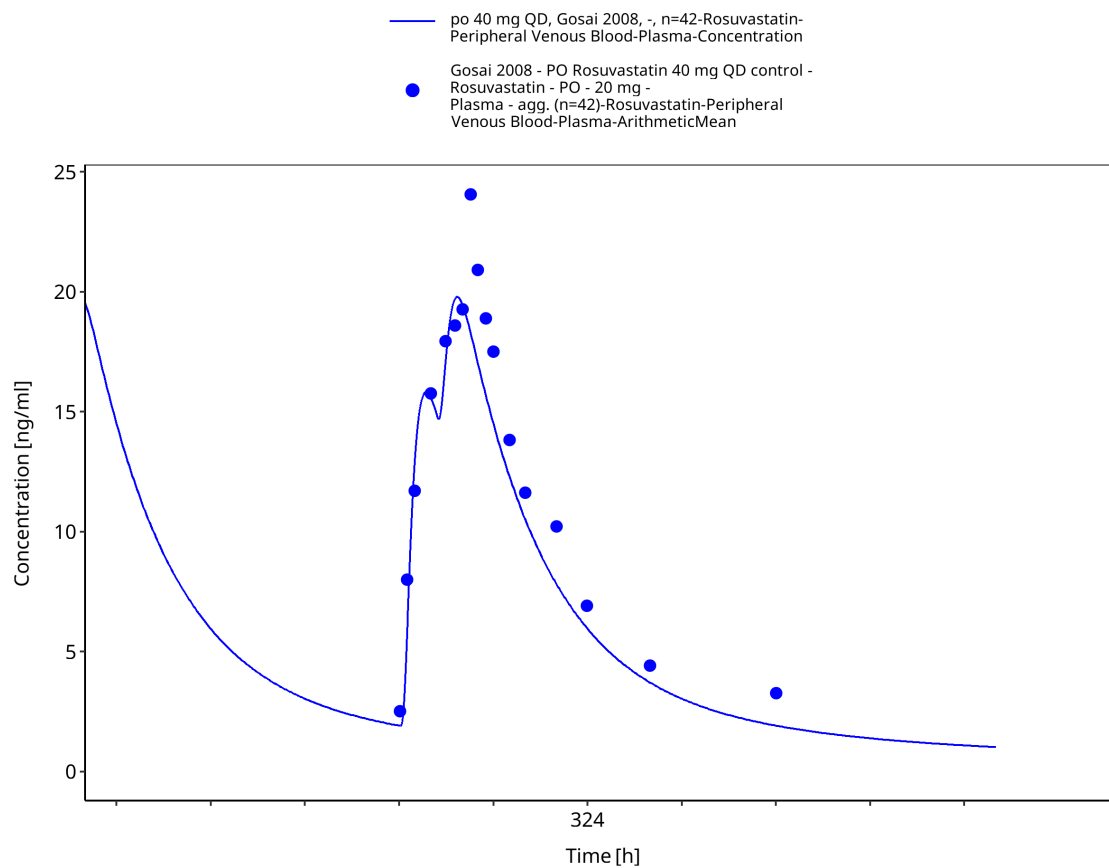


Figure 3-12: Gosai 2008 - PO Rosuvastatin 40 mg QD control

3.3.2 Model Verification

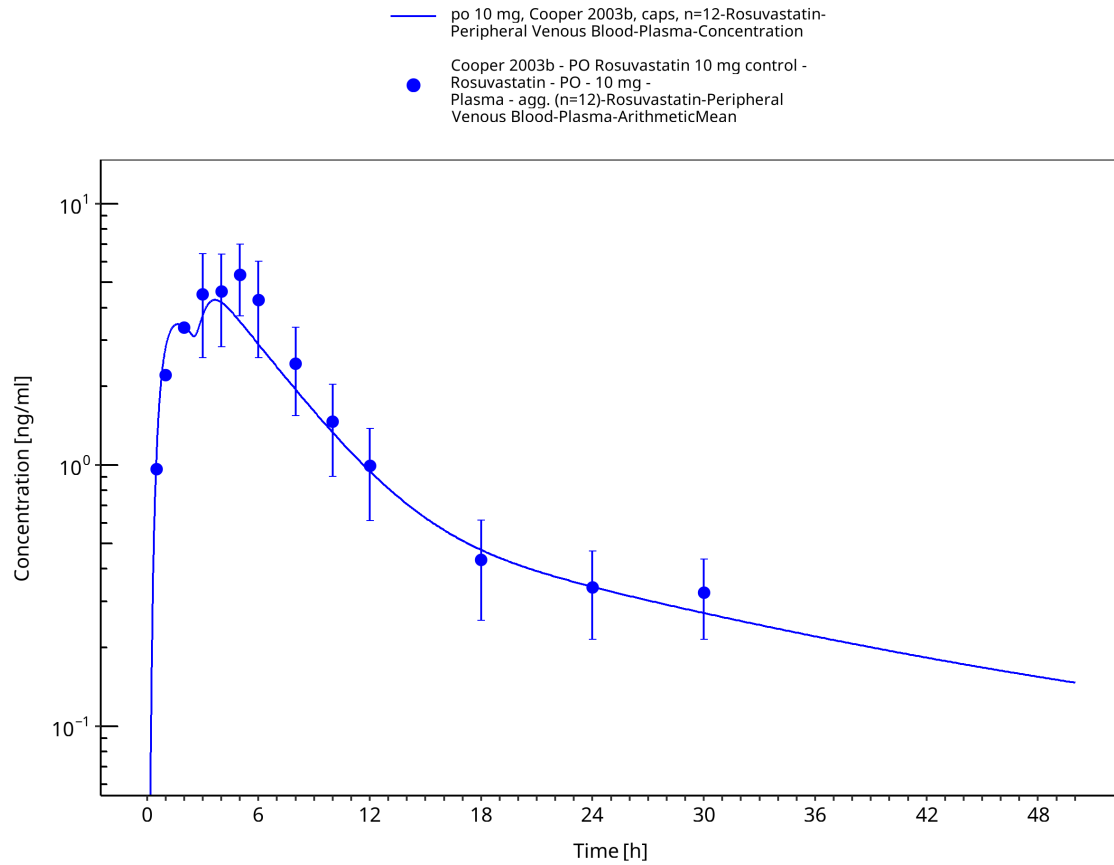


Figure 3-13: po 10 mg, Cooper 2003b, caps, n=12

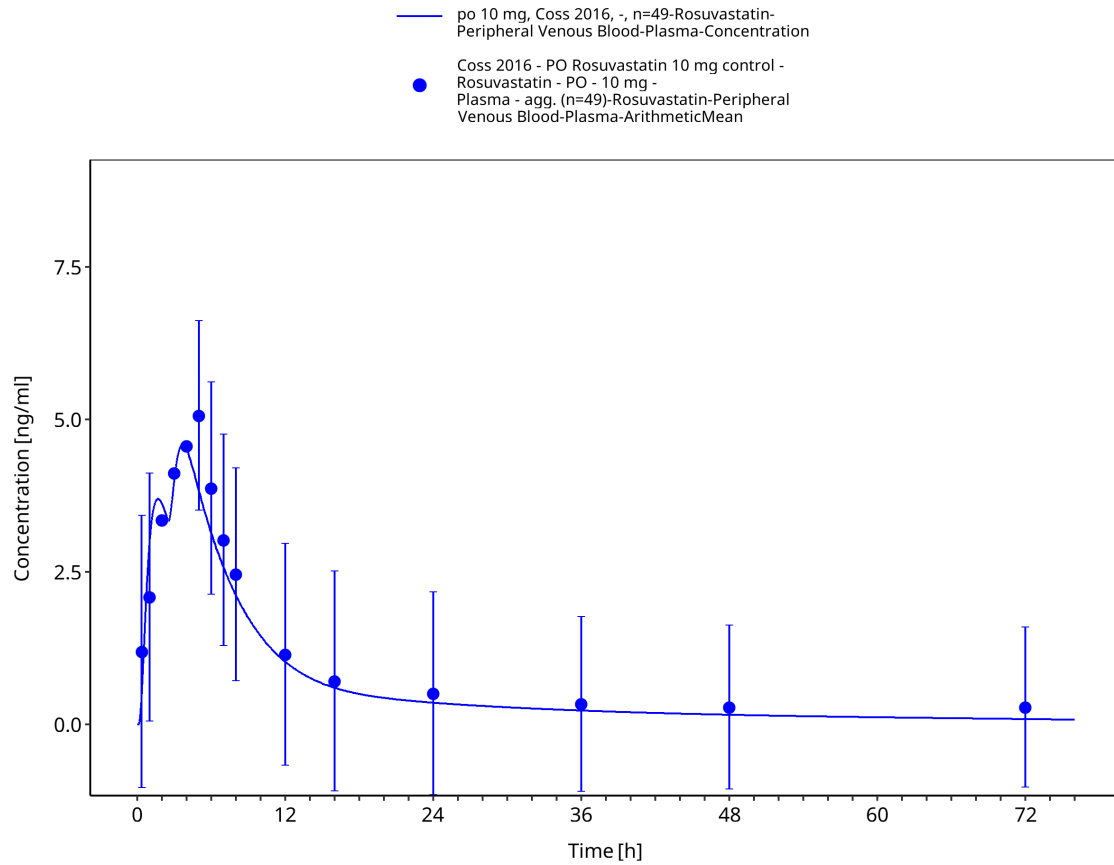


Figure 3-14: Coss 2016 - PO Rosuvastatin 10 mg control

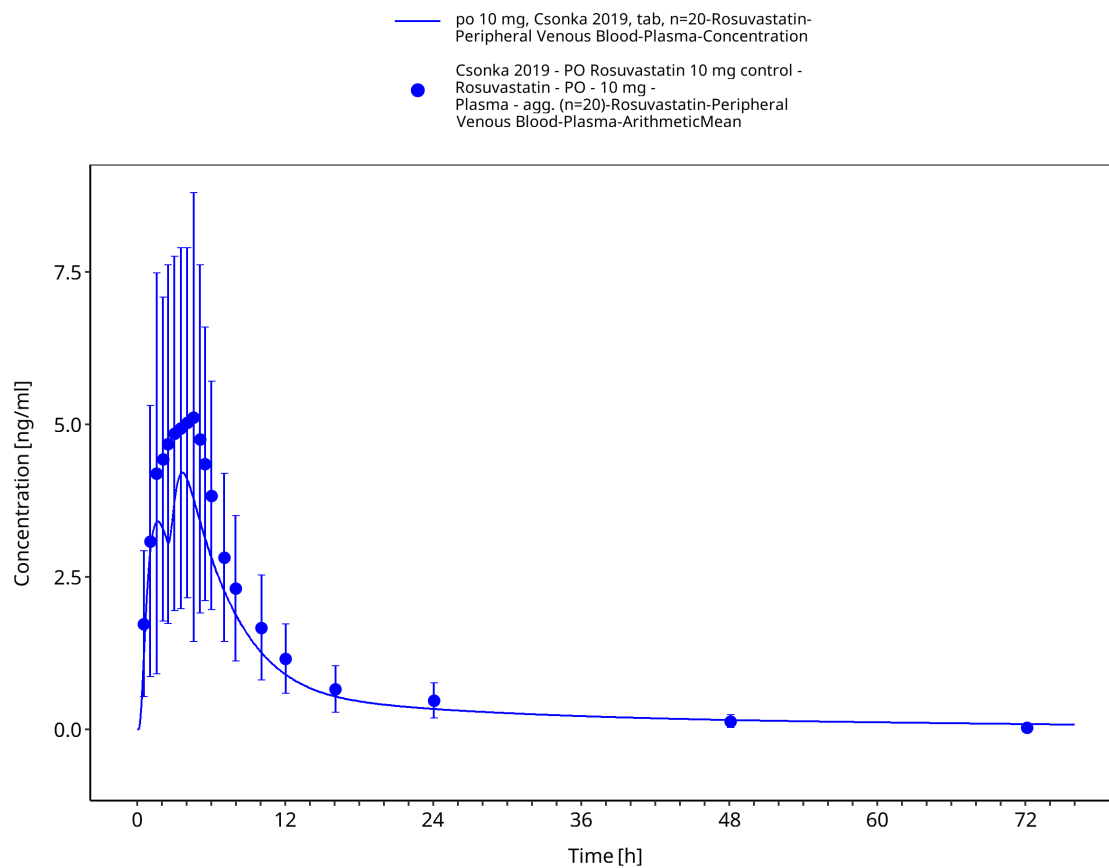


Figure 3-15: Csonka 2019 - PO Rosuvastatin 10 mg control

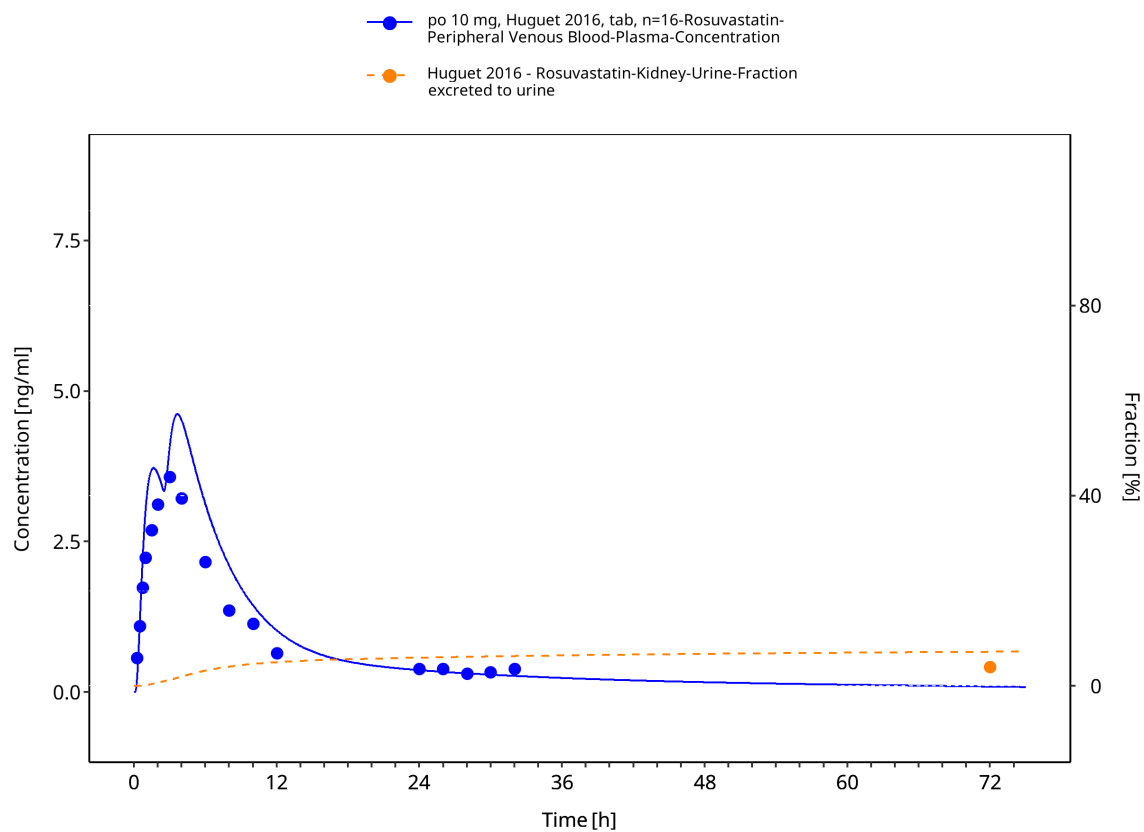


Figure 3-16: Huguet 2016 - PO Rosuvastatin 10 mg control

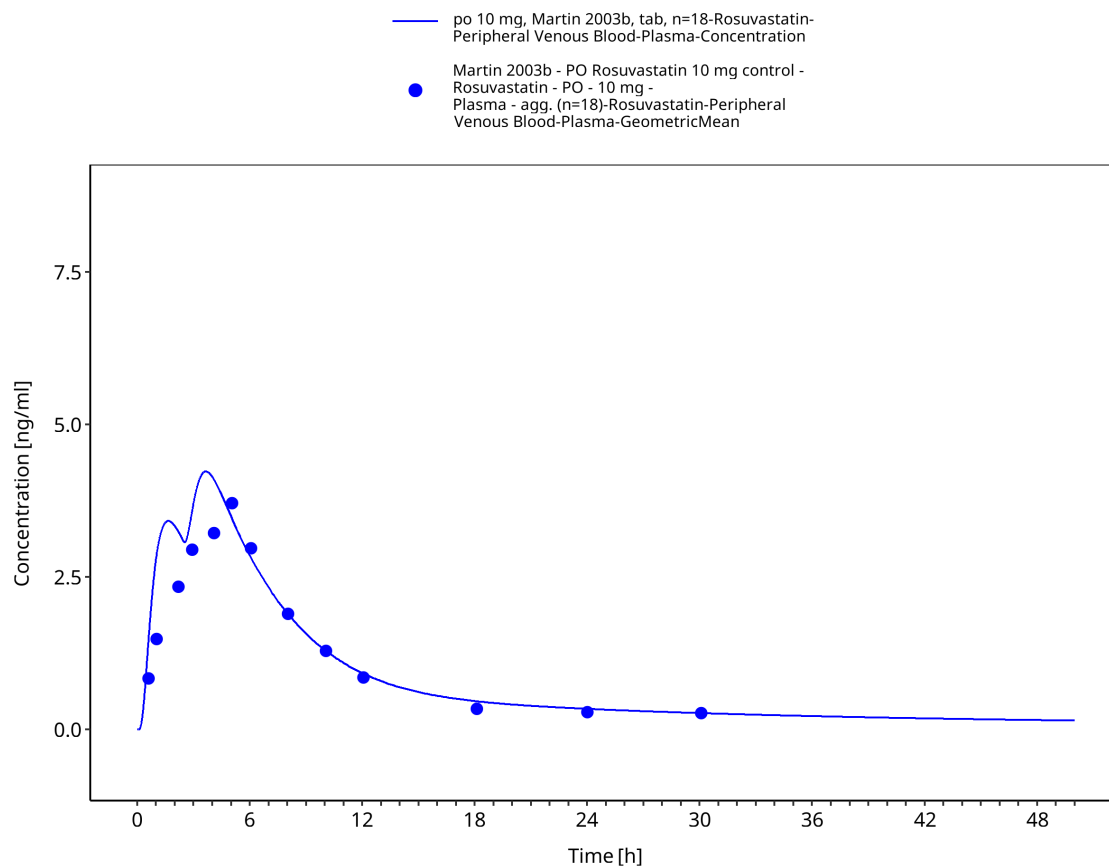


Figure 3-17: Martin 2003b - PO Rosuvastatin 10 mg control

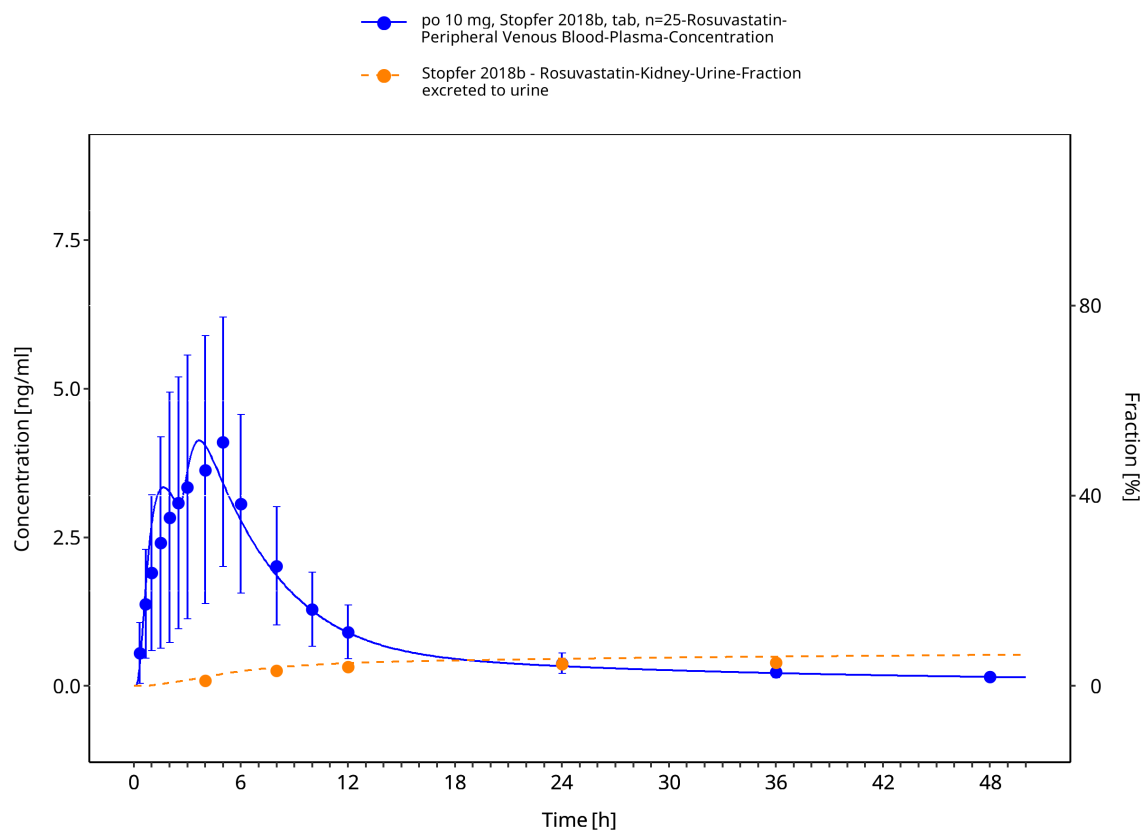


Figure 3-18: Stopfer 2018b - PO Rosuvastatin 10 mg control

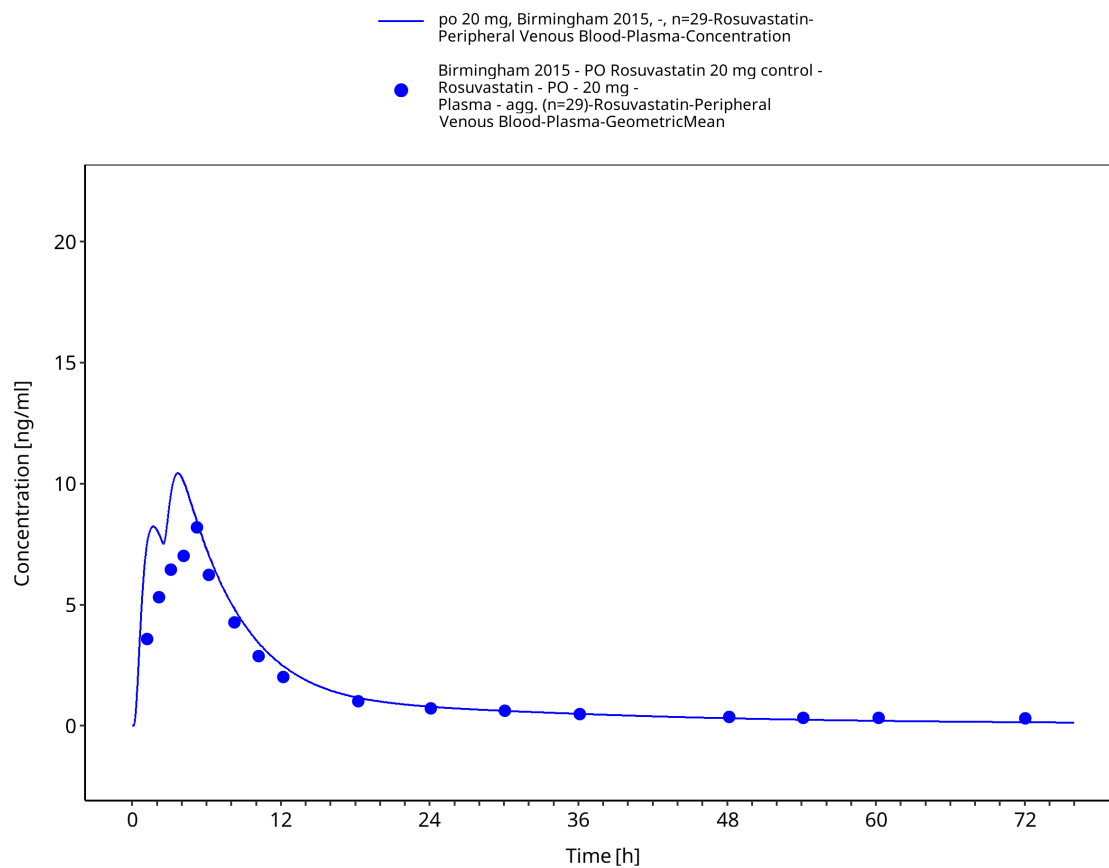


Figure 3-19: Birmingham 2015 - PO Rosuvastatin 20 mg control

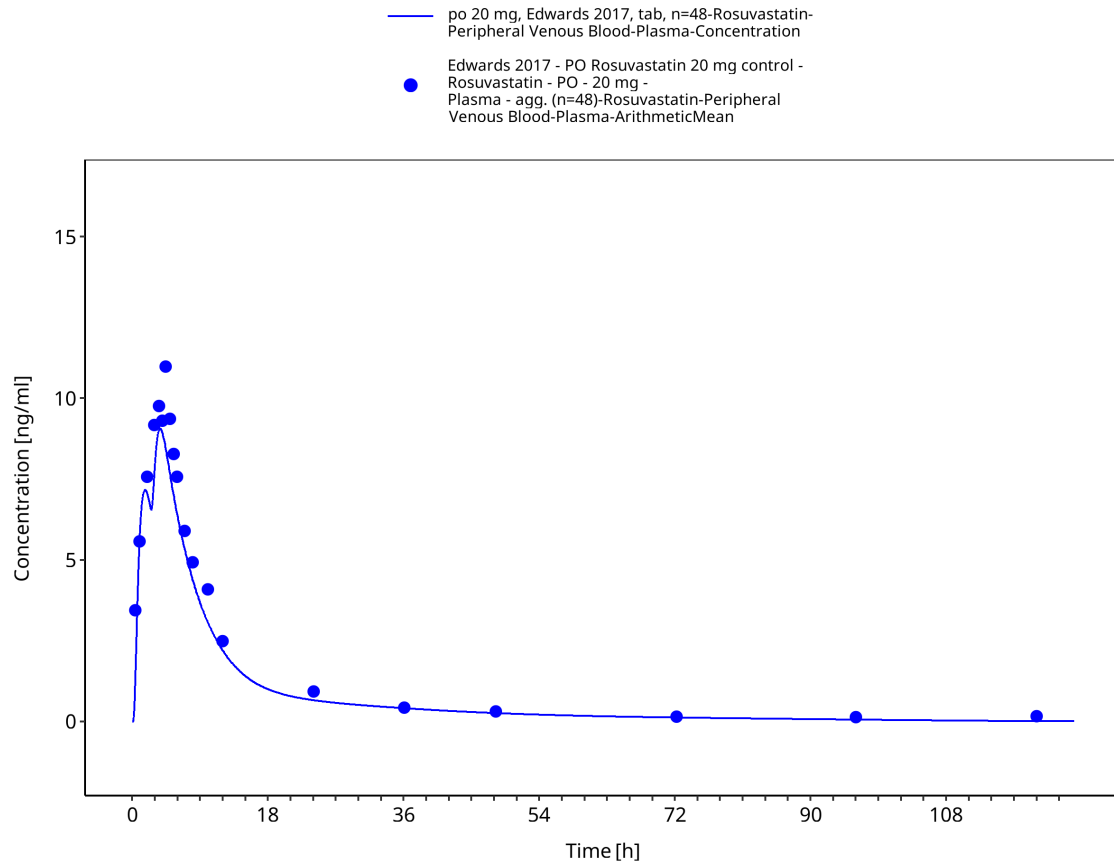


Figure 3-20: Edwards 2017 - PO Rosuvastatin 20 mg control

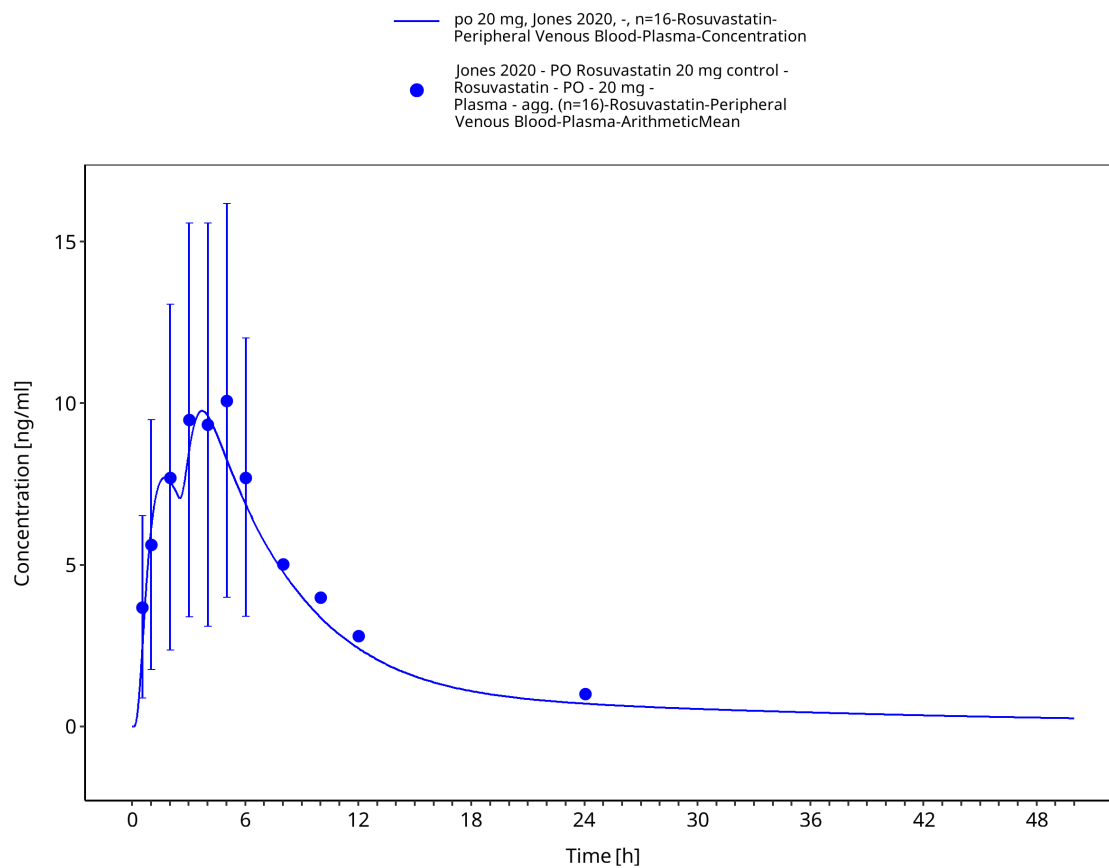


Figure 3-21: Jones 2020 - PO Rosuvastatin 20 mg control

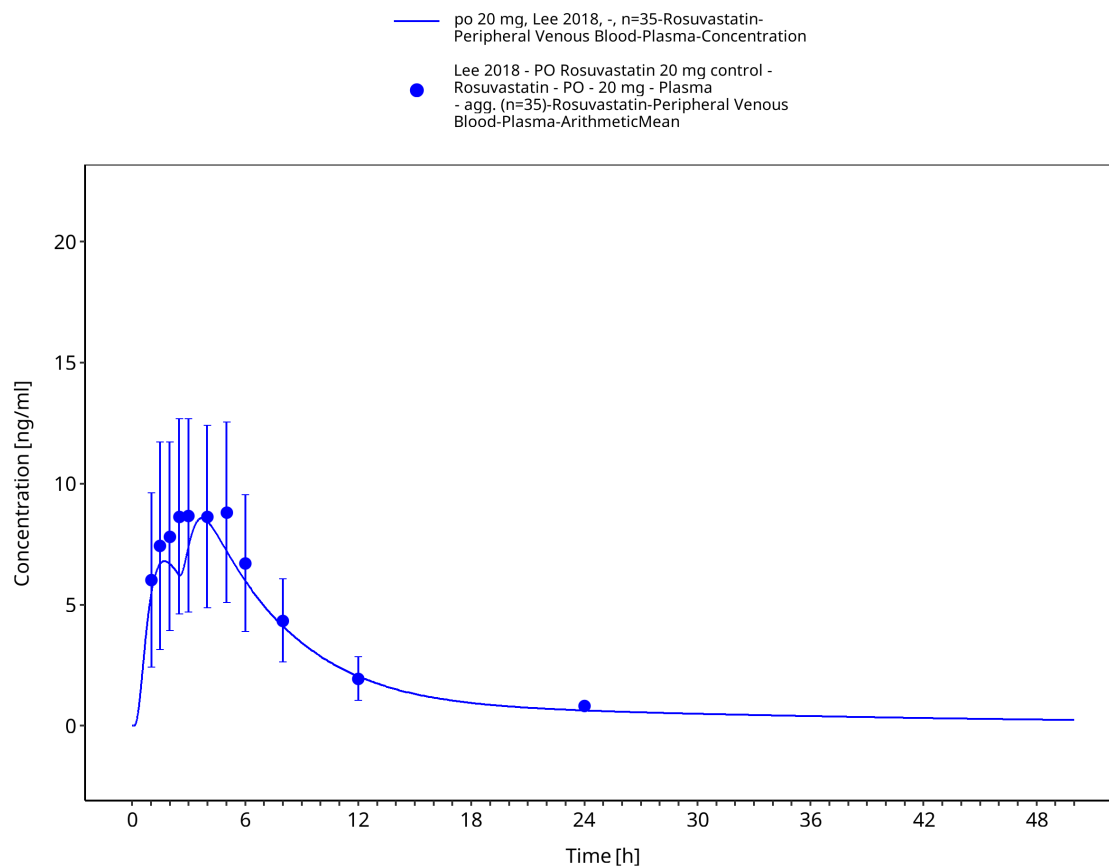


Figure 3-22: Lee 2018 - PO Rosuvastatin 20 mg control

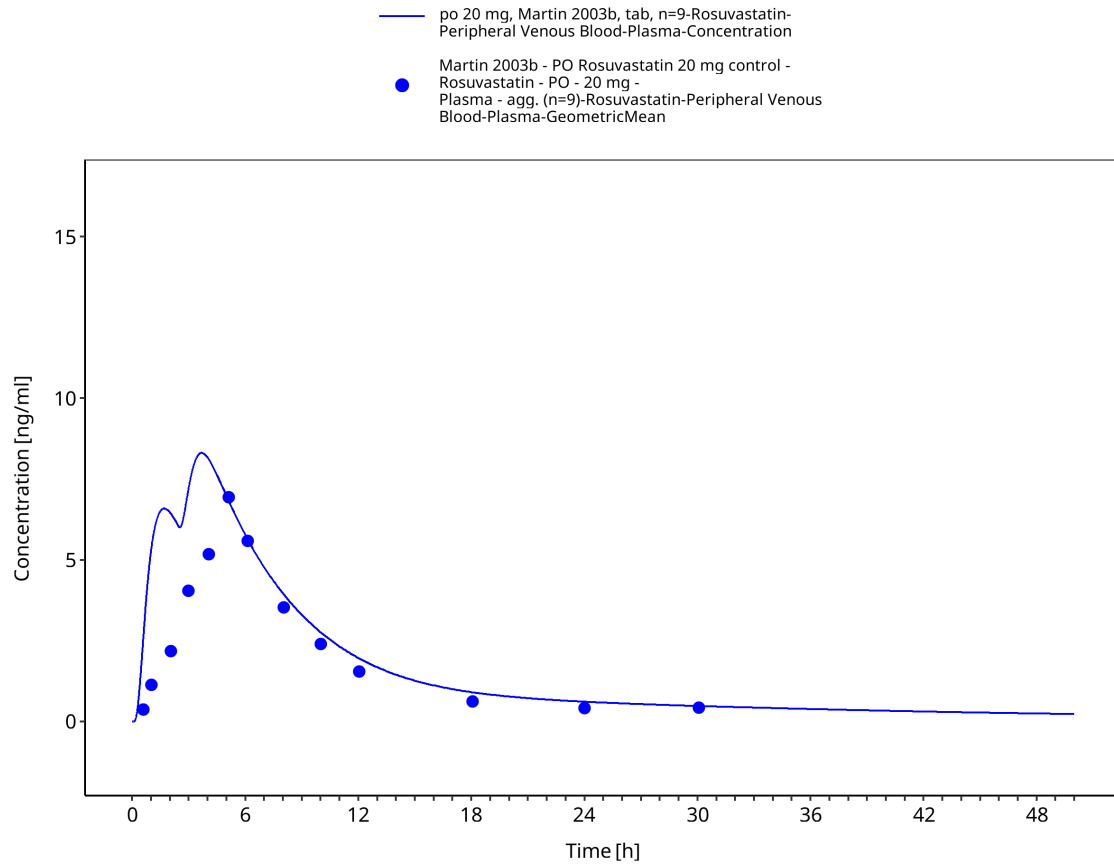


Figure 3-23: Martin 2003b - PO Rosuvastatin 20 mg control

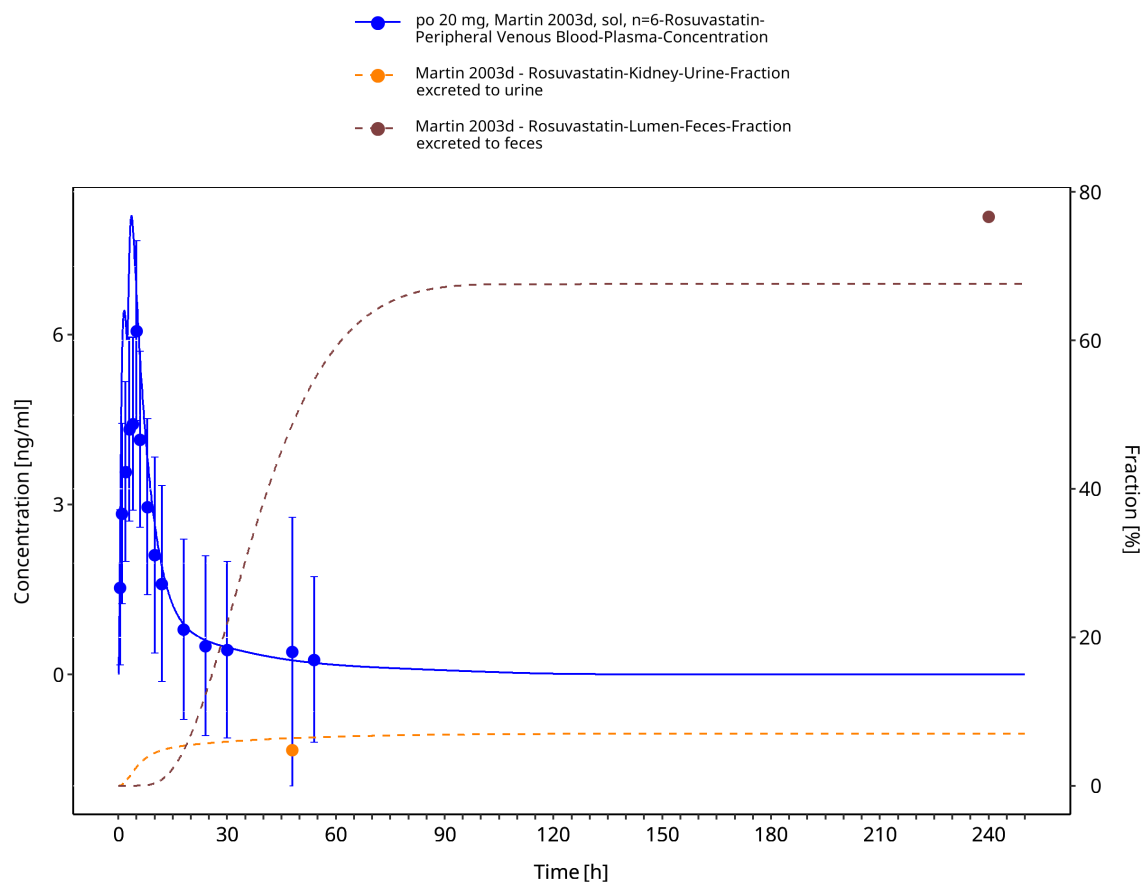


Figure 3-24: Martin 2003d - PO Rosuvastatin 20 mg control

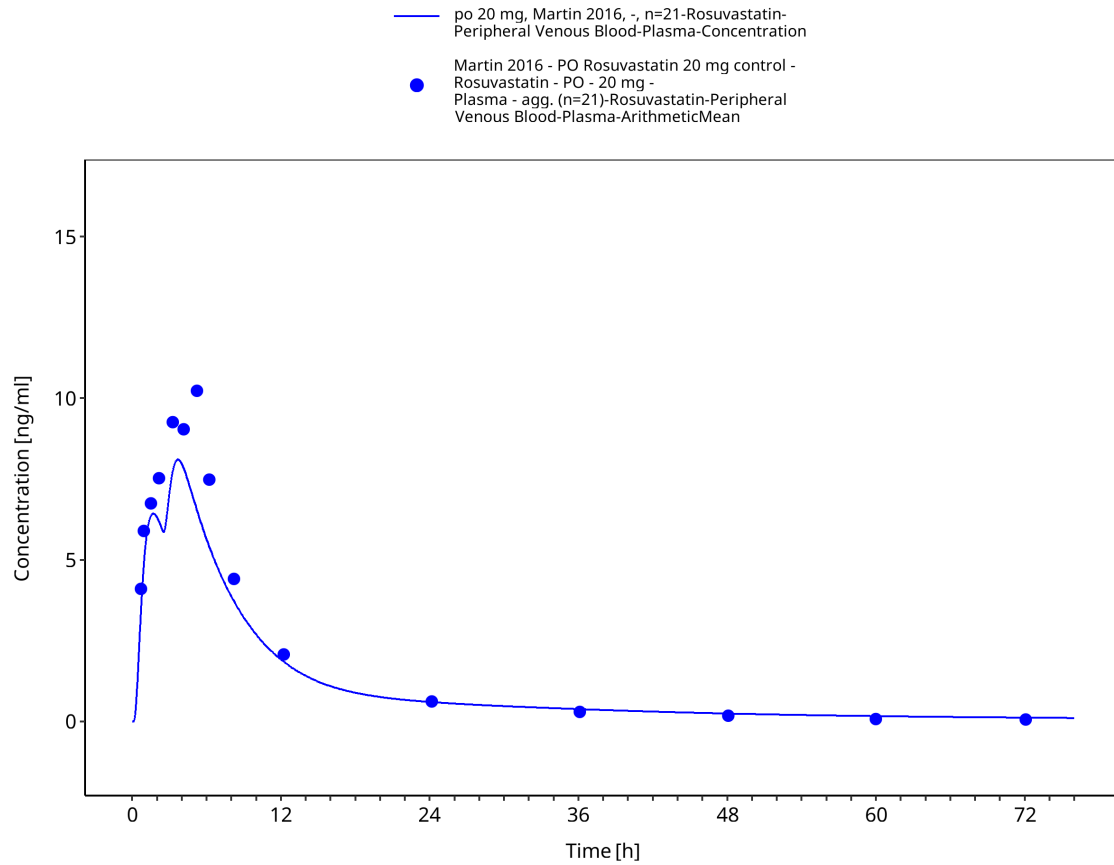


Figure 3-25: Martin 2016 - PO Rosuvastatin 20 mg control

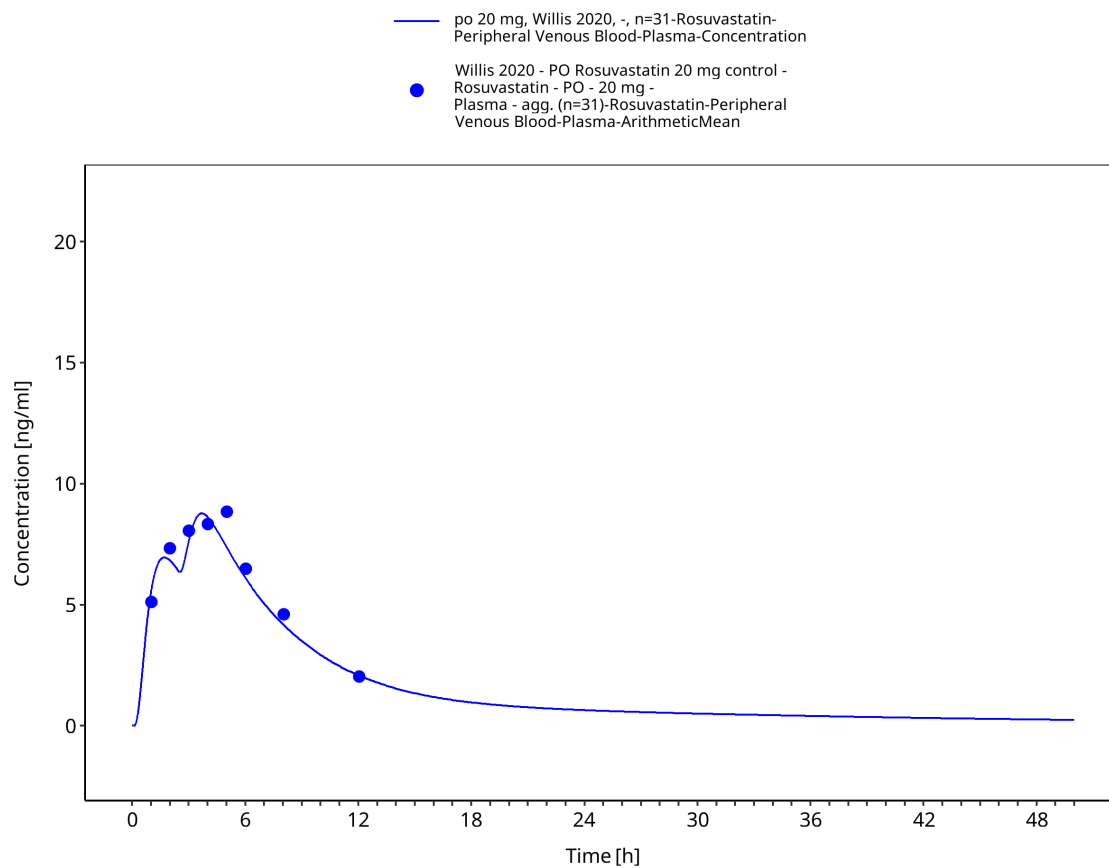


Figure 3-26: Willis 2020 - PO Rosuvastatin 20 mg control

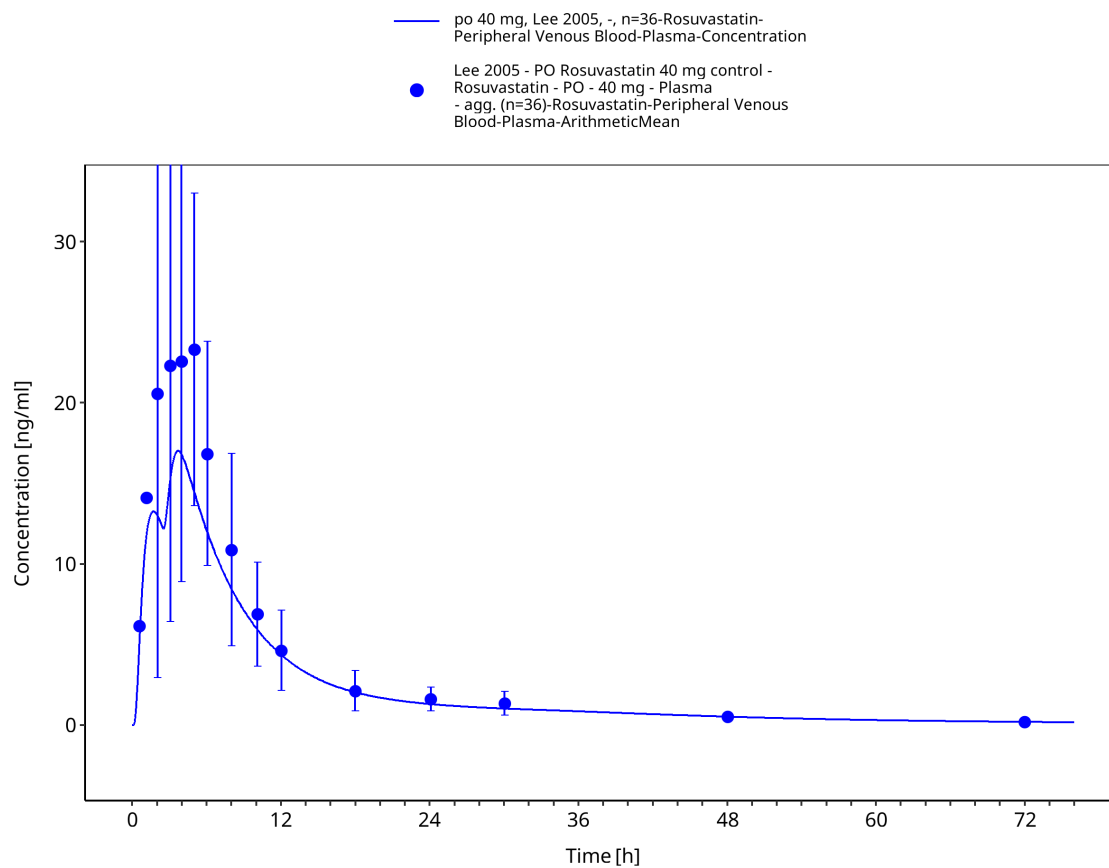


Figure 3-27: Lee 2005 - PO Rosuvastatin 40 mg control

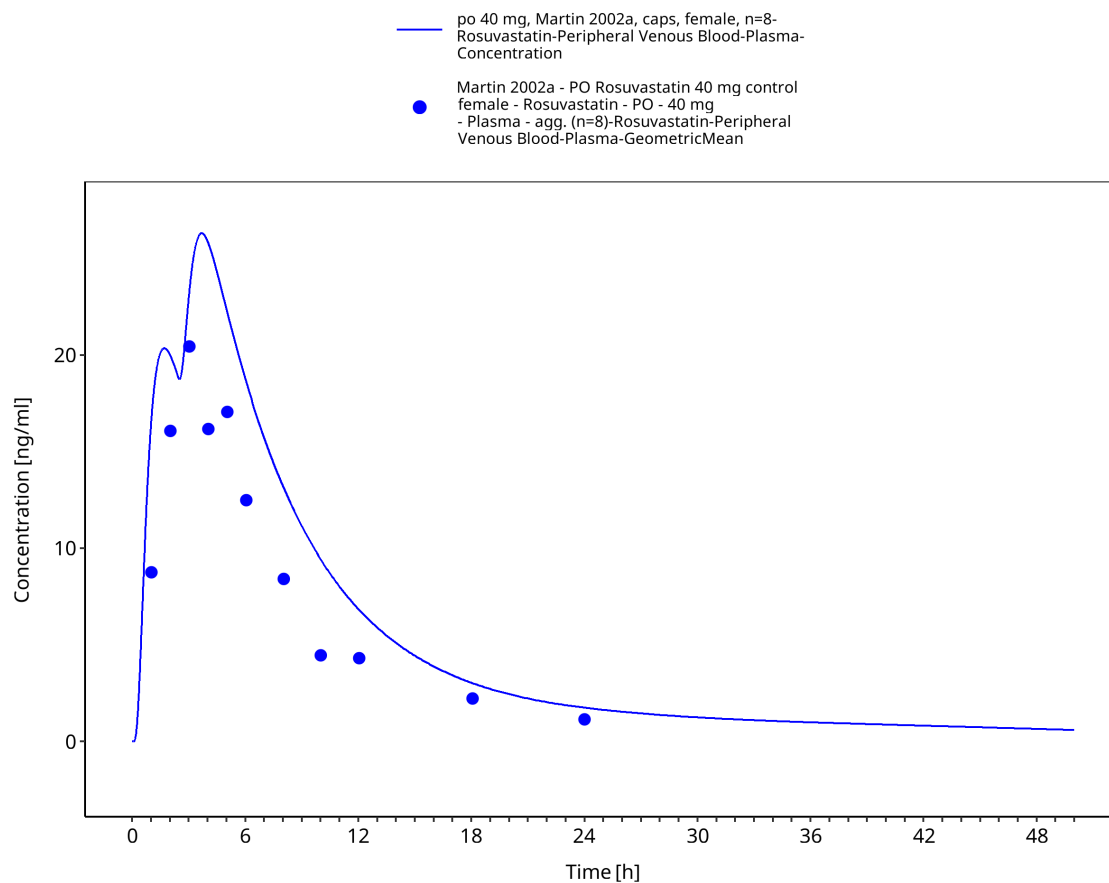


Figure 3-28: Martin 2002a - PO Rosuvastatin 40 mg control female

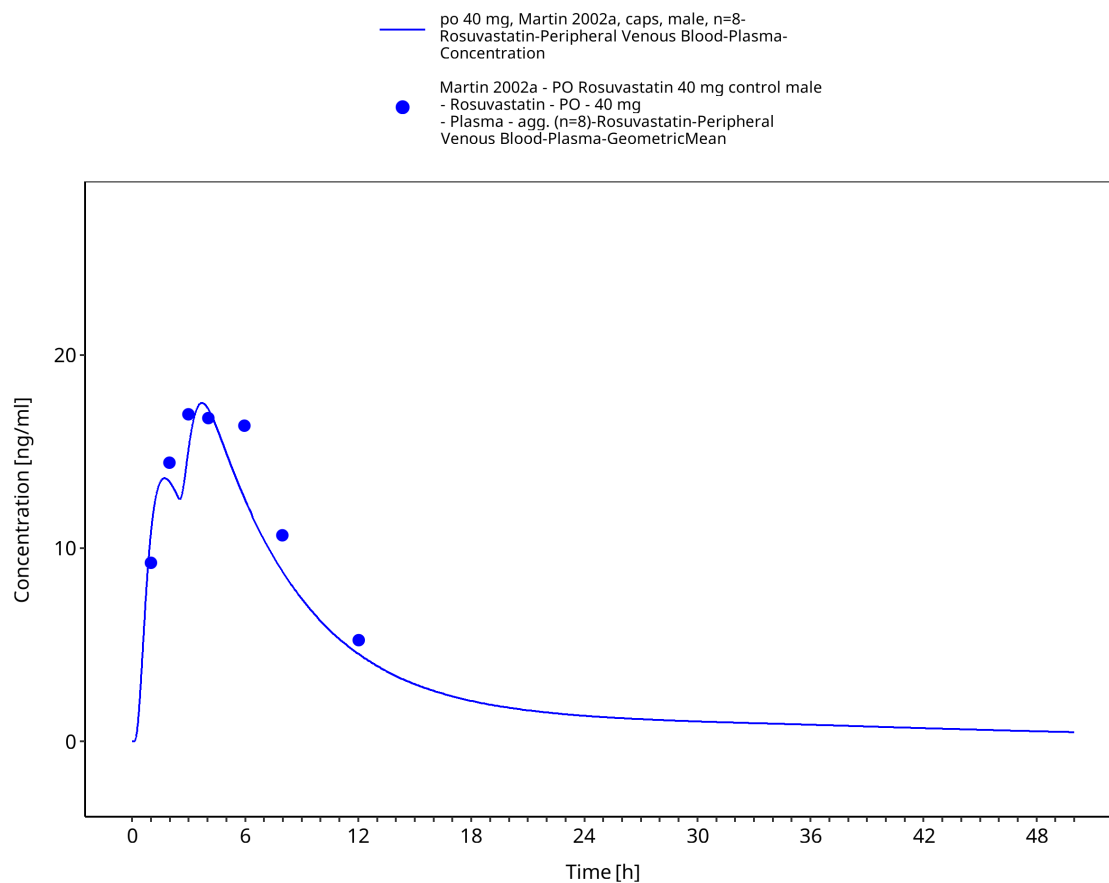


Figure 3-29: Martin 2002a - PO Rosuvastatin 40 mg control male

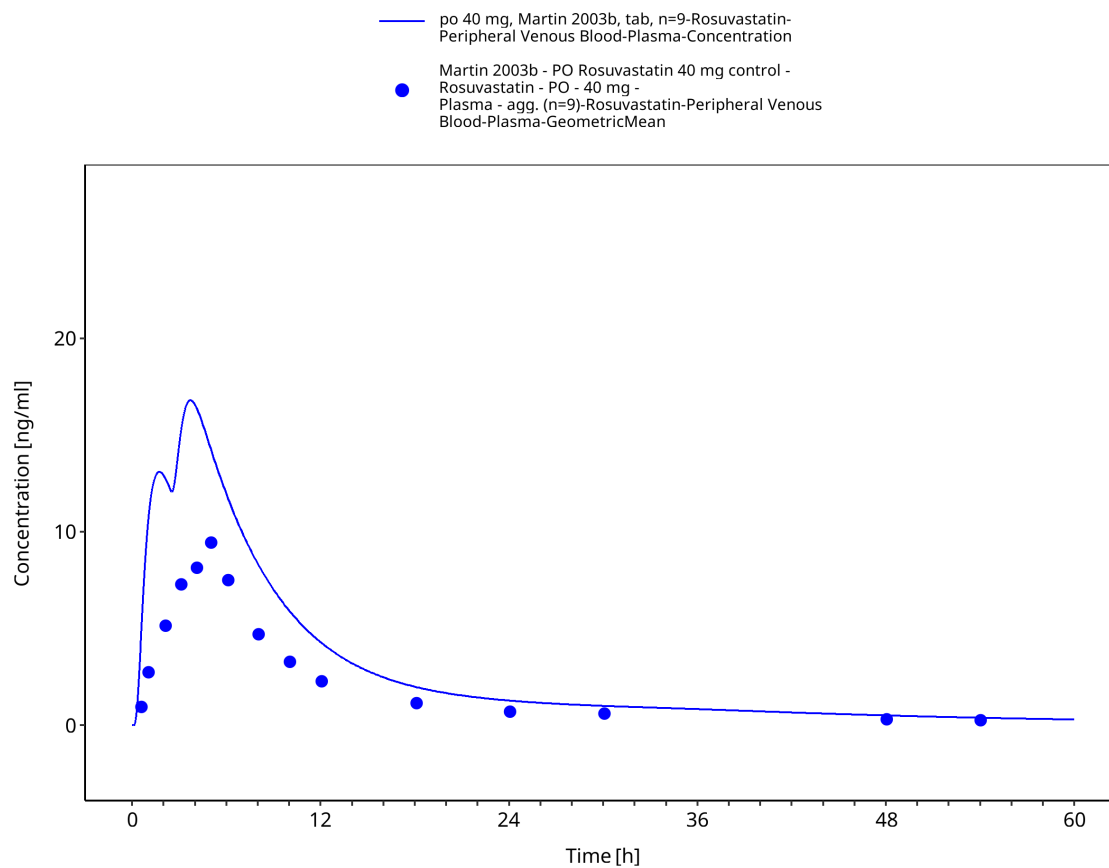


Figure 3-30: po 40 mg, Martin 2003b, tab

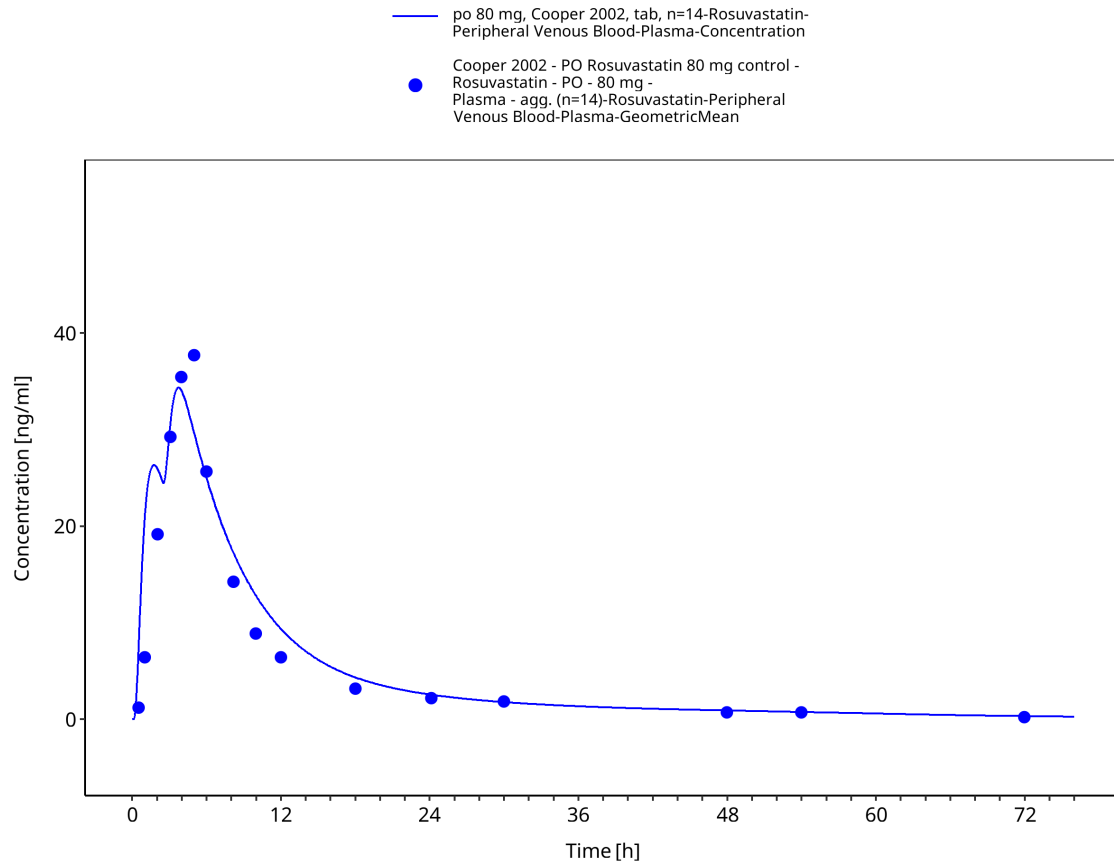


Figure 3-31: Cooper 2002 - PO Rosuvastatin 80 mg control

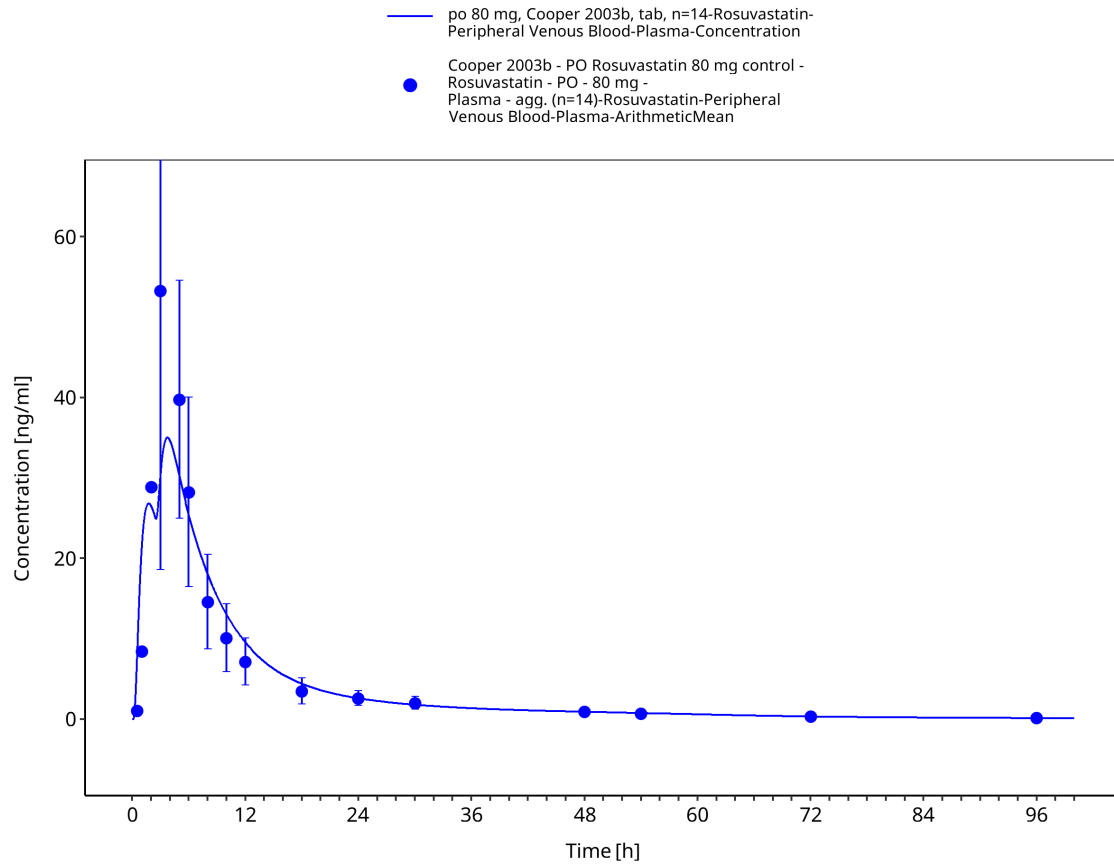


Figure 3-32: Cooper 2003b - PO Rosuvastatin 80 mg control

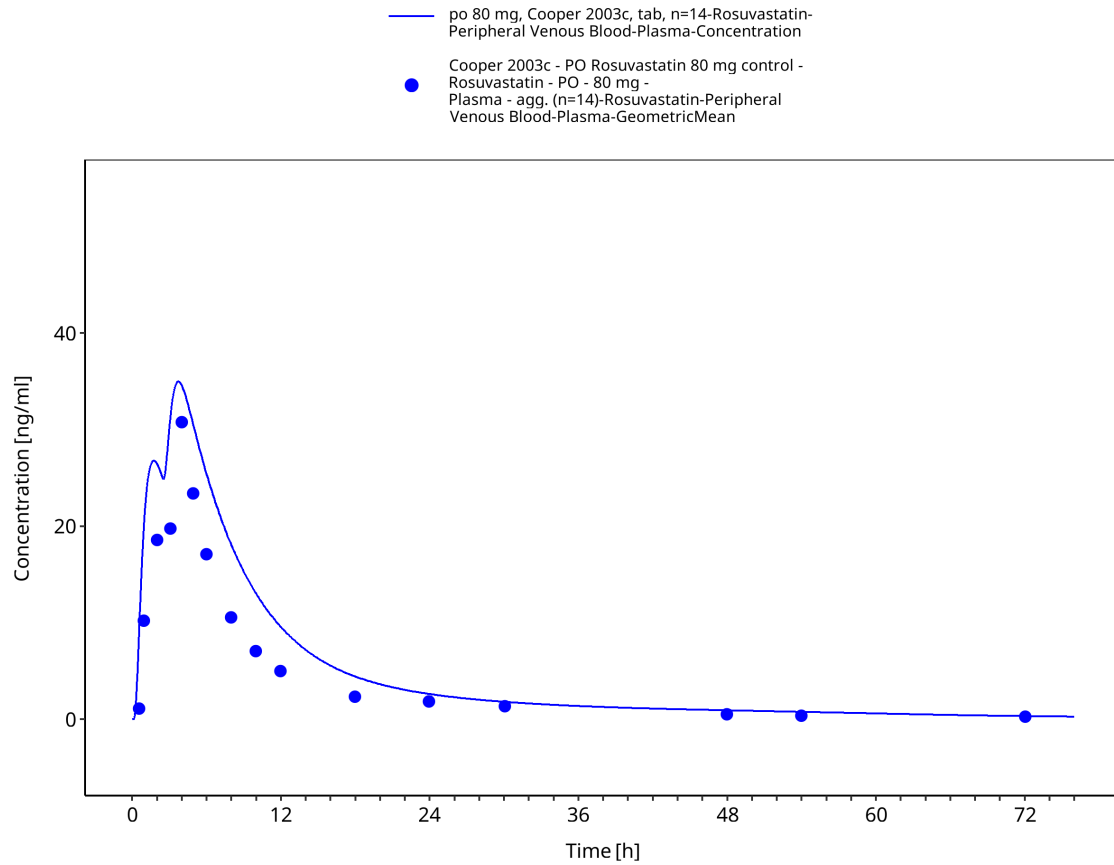


Figure 3-33: Cooper 2003c - PO Rosuvastatin 80 mg control

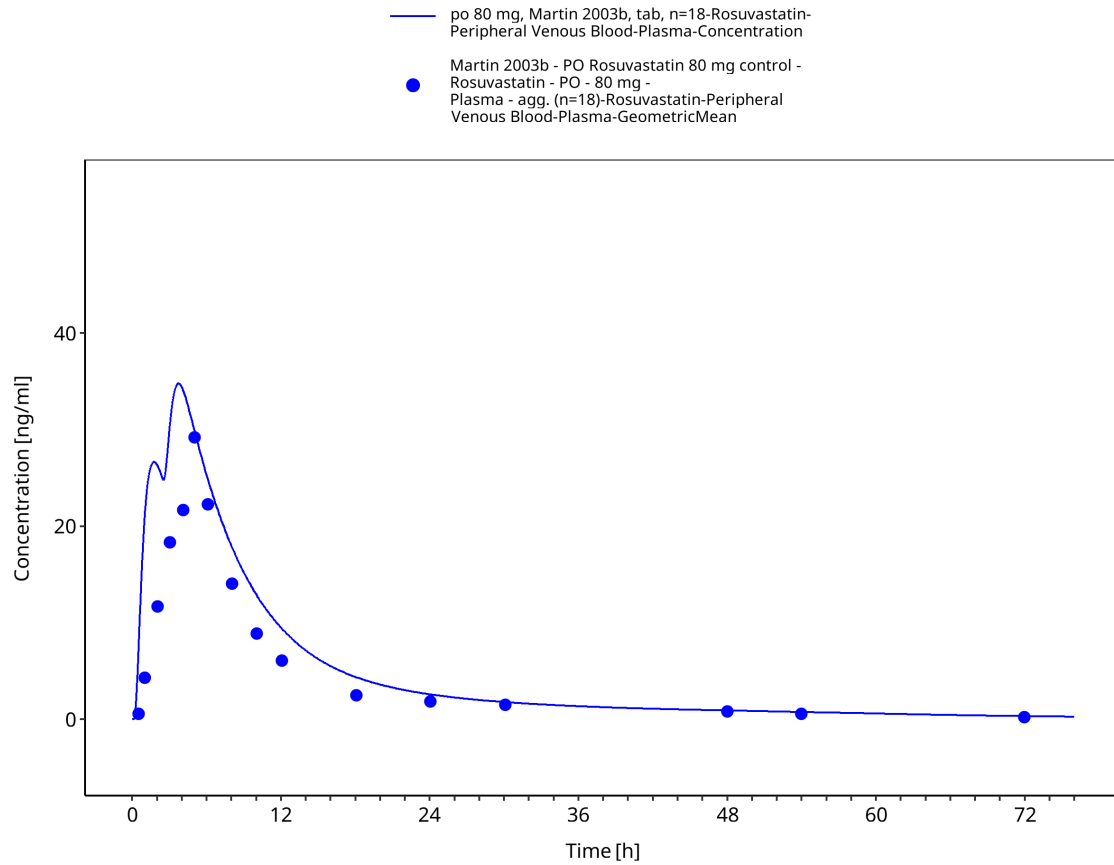


Figure 3-34: Martin 2003b - PO Rosuvastatin 80 mg control

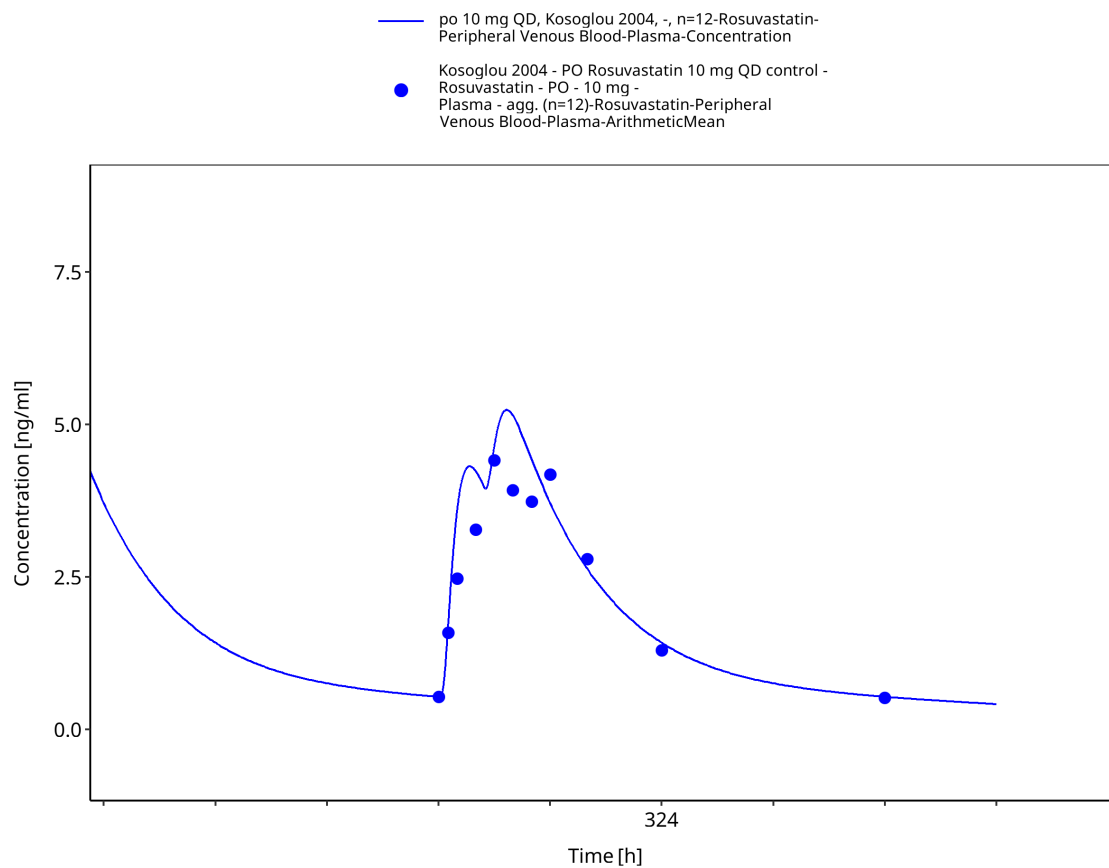


Figure 3-35: Kosoglou 2004 - PO Rosuvastatin 10 mg QD control

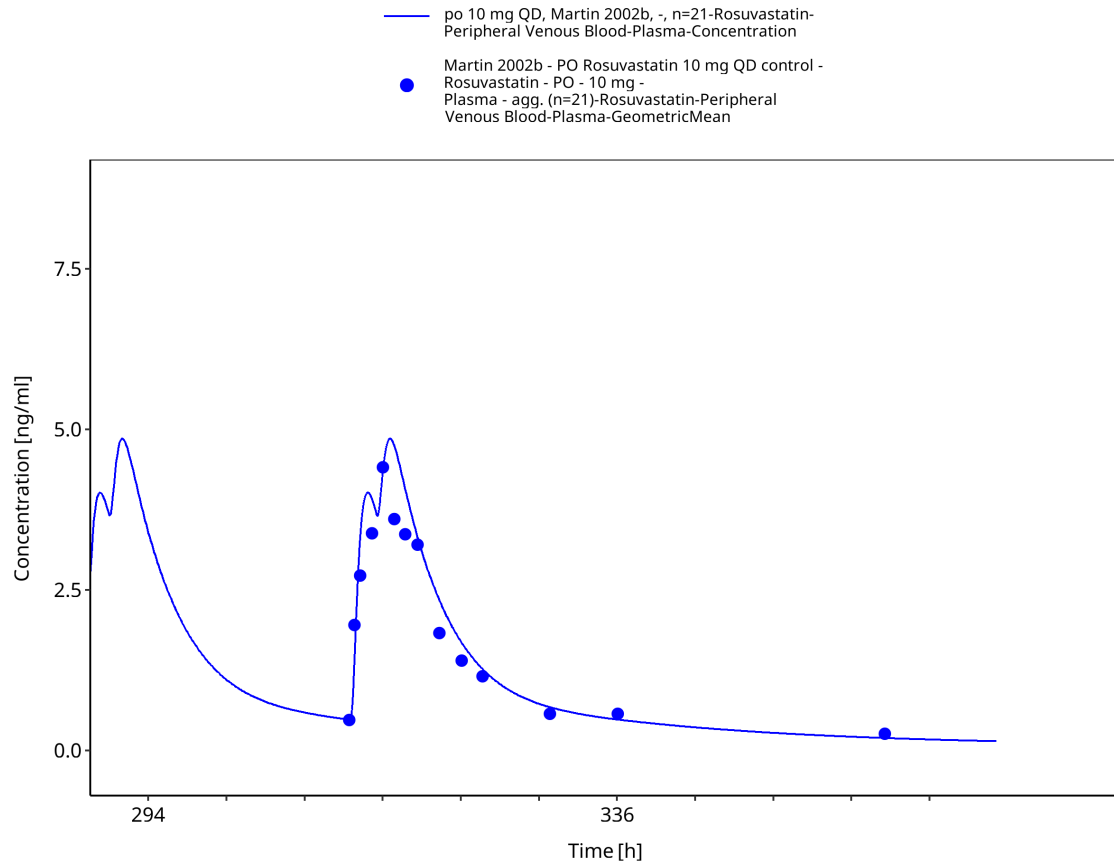


Figure 3-36: Martin 2002b - PO Rosuvastatin 10 mg QD control

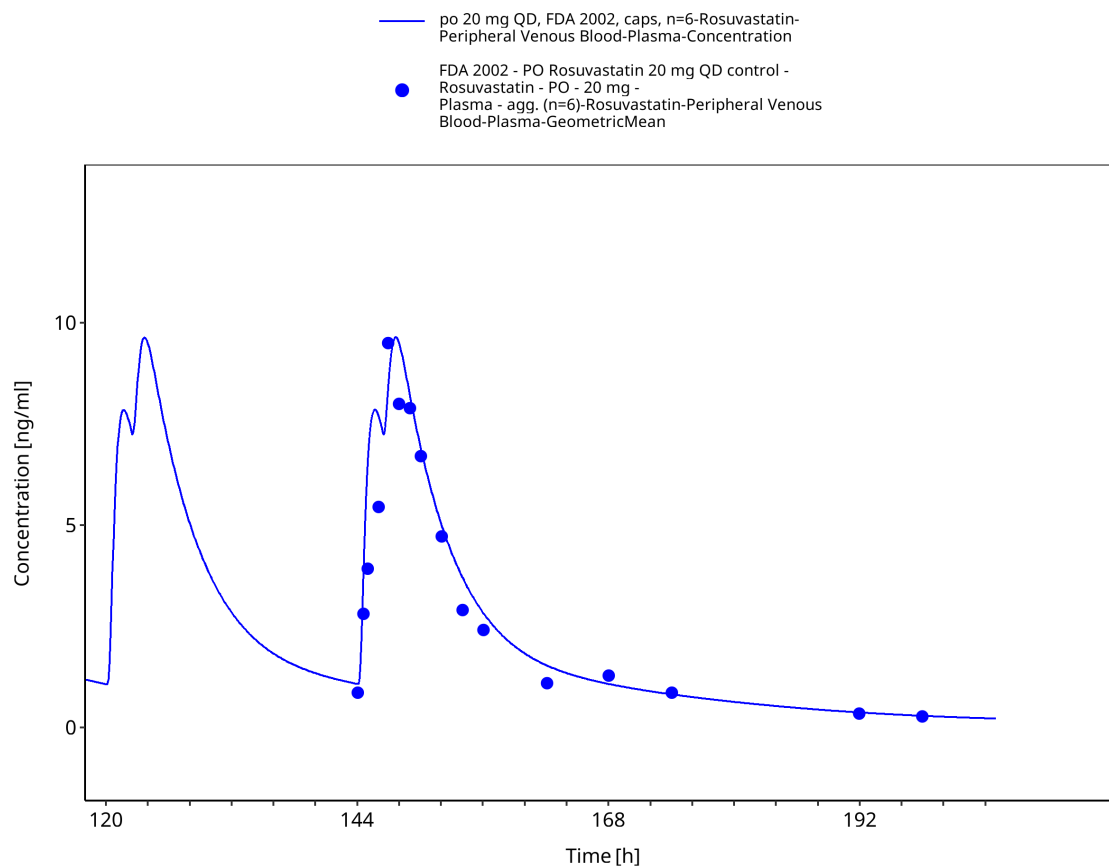


Figure 3-37: FDA 2002 - PO Rosuvastatin 20 mg QD control

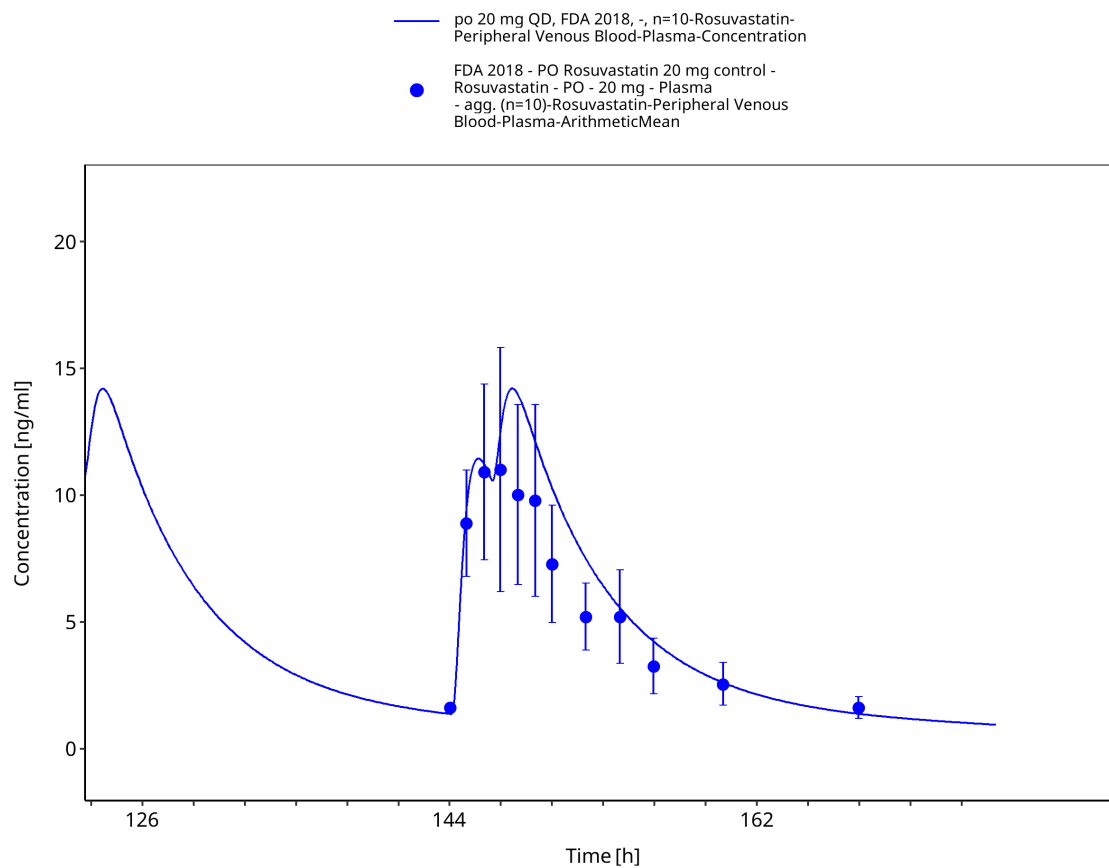


Figure 3-38: FDA 2018 - PO Rosuvastatin 20 mg control

4 Conclusion

The presented PBPK model adequately describes the intravenous and oral pharmacokinetics of rosuvastatin in adults.

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